PRECISION AGRICULTURE

Submitted in partial fulfillment of the requirements of the degree of B.E. in Computer Engineering

B.E. Computer Engineering

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2022 - 2023

CERTIFICATE

This is to certify that **Admon Aloz(A/07), Jaden Butello(A/12), Orvil D'silva(A/30), Basil Koli(A/59), Nigel Lobo(A/60)** are the bonafide students of St. Francis Institute of Technology, Mumbai. They have successfully carried out the project titled "Precision Agriculture" in partial fulfilment of the requirement of B. E. Degree in Computer Engineering of Mumbai University during the academic year 2022-2023. The work has not been presented elsewhere for the award of any other degree or diploma prior to this.

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CERTIFICATE

This is to certify that the project entitled "**Precision Agriculture**" providing smart farming solutions to solve various traditional farming issues for the company Zeptogreens Autonetics Private Limited is a bona fide work of **Basil Koli**, **Orvil D'silva**, **Jaden Butelho**, **Nigel Lobo** and **Admon Aloz**. This group will work on this project for a year as a part of their Final Year (BE) project, meeting the proof of concept requirements and objectives for us.

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Figure 1: Certificate

Project Report Approval for B.E.

This project entitled ***'Precision Agriculture'*** by **Admon Aloz, Jaden Butello, Orvil D'silva, Basil Koli, Nigel Lobo** is approved for the degree of Bachelor of Engineering in Computer from University of Mumbai.

Examiners

1. - - - - - - - - - - - - - - - - - -

2. - - - - - - - - - - - - - - - - - -

Date: 04/05/2023

Place: Mumbai

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included; we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in this submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Signatures of all the students in the group

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Abstract

Agriculture is one of the top three contributors to the Indian economy. However, most of the agricultural practices are still quite traditional in nature. During these times, when the world is facing very intense population growth, farmers who are striving to meet the global food demands have become true heroes. Unfortunately, their mission of feeding the world is a demanding battle with unfavorable weather conditions, various pests, weeds and plant disease that work against them. Precision agriculture management practices can significantly help mitigate these problems while boosting yields. Precision Agriculture is a Mobile or Web application which improves the efficiency of agricultural activities via minimal initial input of material and human resources and avoiding harmful effects on the environment on one hand and automatizing the production on another hand, thus providing environmental, social and economic benefits. Farmers are able to get insights of their farm, by using our system‘s modules such as Disease Detection and Classification, Species Recognition, Pest Prediction and Classification, Weed Classification, Crop Recommendation and Crop Yield Prediction. For instance, for Crop Recommendation, the classification is done based on the values from the soil's report, and numeric data is required as input for Crop Yield Prediction. On the contrary, for Species Recognition or Disease Detection and Classification, images are taken as input.

Therefore, the system can be divided into two modules where one requires image data as input and the other requires numeric data as input. Various datasets were collected for each module which include the PlantVillage dataset, the Pest Dataset etc. For image input based modules like pest prediction and classification, the input image from the dataset is preprocessed and segmented using techniques like Otsu‘s Thresholding and Morphological Transform, followed by extracting texture features using Convolution Neural Networks (CNN). We have also calculated color and texture features which also equally play a pivotal role in feature extraction to perform Bins Approach for classification. The algorithms which were used for image input based modules were CNN with ResNet50 and DenseNet and Bins Approach with Support Vector Machine (SVM), Decision Tree and Logistic Regression. And the algorithms used for numeric data based modules were Naive Bayes, Decision

Tree, Random Forest, SVM and XGBRegressor. Performance of the image modules and Crop Recommendation module is evaluated and validated using the following parameters: Accuracy, Precision, Recall and the F1-score. Performance of Crop Yield Prediction is evaluated and validated using the following parameters: MAE, MSE and R2.

Keywords: *Precision Agriculture, Disease Detection and Classification, Species recognition, Pests Prediction and Classification, Weed Classification, CNN, ResNet50, Densenet, Bins Approach, SVM, Logistic Regression, Decision Tree, Naive Bayes, Random Forest, XGBRegressor, Accuracy, Precision, Recall, F1-Score, MAE, MSE, R2*

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Chapter 1

Introduction

It is expected that by 2050, the global population will reach about 9.6 billion, and food production must effectively double from current levels in order to feed every mouth. Agri-technology and precision farming have arisen as new scientific fields that use data intensive approaches to drive agricultural productivity. With new technological advancements in the agricultural revolution of precision farming, each farmer can be able to produce enough resources to meet the requirement. The first wave of the precision agricultural revolution came in the forms of satellite and aerial imagery, weather prediction, variable rate fertilizer application, and crop health indicators. The second wave aggregates the machine data for even more precise planting, topographical mapping, and soil data. Precision agriculture is a key component of the third wave of modern agricultural revolutions.

1.1 Description

Agriculture is one of the top three contributors to the Indian economy. However, most of the agricultural practices are still quite traditional in nature. Precision agriculture management practices can significantly reduce the amount of nutrient and other crop inputs used while boosting yields. Farmers thus obtain a return on their investment by saving on water, pesticide, and fertilizer costs.

Precision agriculture is the science of improving crop yields and assisting management decisions using high technology sensor and analysis tools.It is a new concept adopted throughout the world to increase production, reduce labor time, and ensure the effective management of fertilizers and irrigation processes. It uses a large amount of data and information to improve the use of agricultural resources, yields, and the quality of crops.

Precision agriculture is an advanced innovation and optimized field level management strategy used in agriculture that aims to improve the productivity of resources on agriculture fields.It has become a cornerstone of sustainable agriculture, since it respects crops, soils and farmers. Sustainable agriculture seeks to assure a continued supply of food within the ecological, economic and social limits required to sustain production in the long term.

1.2 Problem Formulation

During these times, when the world is facing very intense population growth, farmers who are striving to meet the global food demands have become true heroes. Unfortunately, their mission of feeding the world is a demanding battle with unfavourable weather conditions, various pests, weeds and plant disease that work against them.

Modern agricultural production relies on monitoring crop status by observing and measuring variables such as soil condition, plant health, fertilizer and pesticide effect, irrigation, and crop yield. Managing all of these factors is a considerable challenge for crop producers. The rapid enhancement of precise monitoring of agricultural growth and its health assessment is important for sensible use of farming resources as well as in managing crop yields. Nowadays, Increased Land degradation is also a major problem.Out of million ha about 144 million ha of land are affected by water or wind erosion alone.

This also leads to massive depletion of Water resources. Also at present, there is an increasing commitment to reduce reliance on excessive chemical inputs in agriculture. Numerous technologies have been applied to make agricultural products safer and to lower their adverse impacts on the environment, a goal that is consistent with sustainable agriculture. Precision agriculture has emerged as a valuable component of the framework to achieve this goal.

1.3 Motivation

Precision agriculture is an innovative, integrated and internationally standardized approach aiming to increase the efficiency of resource use and to reduce the uncertainty of decisions required to manage variability on farms. It has been hailed as one of the most scientific and modern approaches to production agriculture in the 21st century, as it epitomizes a better balance between reliance on traditional knowledge and information and management-intensive technologies.

For the past several decades, agricultural technology has made tremendous strides in fostering increased crop yields and more efficient use of inputs such as water and fertilizer. As climate change risks are recognized, increasing uncertainties cloud the outlook for the continuation of historical trends in the growth of crop yields. Success in securing sustainable food will be highly dependent on the deployment of Precision agriculture technologies to improve the efficiency and resiliency of agricultural systems.

1.4 Proposed Solution

Precision Agriculture will improve the efficiency of agricultural activities via minimal initial input of material and human resources and avoiding harmful effects on the environment on one hand and automatizing the production on another hand, thus providing environmental, social and economic benefits.

Precision Agriculture will assist farmers with the insights on their field status while providing suitable recommendations to overcome complications with regards to field health.

Features:

- Predicting crop yield and based on that deciding what nutrients are to be provided to crops.
- Crop recommendation based on the parameters like pH, temperature, humidity etc.
- Detecting and classifying diseases, weeds, pests and crop species.

The system can be divided into two modules where one requires image data as input and the other requires numeric data as input. Various datasets were collected for each module which include the PlantVillage dataset, the Pest Dataset etc. For image input based modules like pest prediction and classification, the input image from the dataset is preprocessed and segmented using techniques like Otsu‘s Thresholding and Morphological Transform, followed by extracting texture features using Convolution Neural Networks (CNN). We have also calculated color and texture features which also equally play a pivotal role in feature extraction to perform Bins Approach for classification. The algorithms which were used for image input based modules were CNN with ResNet50 and DenseNet and Bins Approach with Support Vector Machine (SVM), Decision Tree and Logistic Regression. And the algorithms used for numeric data based modules were Naive Bayes, Decision Tree, Random Forest and SVM. Performance of the system is evaluated and validated using the following parameters: Accuracy, Precision, Recall and the F1-score.

1.5 Scope of the Project

Precision Agriculture will enhance agricultural productivity and prevent soil degradation in cultivable land resulting in sustained agricultural development. It will reduce excessive chemical usage in crop production. Water resources will be utilized efficiently under the precision farming. It will minimize the risk to the environment particularly with respect to the nitrate leaching and groundwater contamination by means of the optimization of agro-chemical products. Refinement and wider application of precision agriculture technologies in India can help in reducing production costs, increasing productivity and better utilization of natural resources. It has the ability to revolutionize modern farm management in India through improvement in profitability, productivity, sustainability, crop quality, environmental protection, on-farm quality of life, food safety and rural economic development. It will provide opportunities for better resource management and hence reduce wastage of resources

Chapter 2

Literature Review

Weed Detection and Identification

[11] In this paper, a drone is used to collect data so that the WIS can determine if there exist weed or not. The operations of the WIS can be divided into three parts: data collection, data preprocessing and sample generation, and weed identification. The images collected are then partitioned into multiple sub-images such that a single plant is located at the center of a sub-image. It then checks each sub-image to check if there exist weed or not. The WIS functions using the RGB images of the plant as training data. CNN is used for image processing as it effectively reduces image size while preserving the features. It was observed that using histogram equalization is better than using local binary pattern for identification of crops/weeds. The experimental result states that using only machine learning model CNN or only image processing is not good enough in identifying weeds. The identification of weed and crops can be classified as a fine-grained visual classification problem, i.e., to classify similar objects. Using only RGB images to classify crops and weed is a challenging task and is prone to error.

The approach proposed in this paper for identification of weeds is composed of two stages [7]. The first stage consists of the state-of-art CenterNet algorithm for detecting bok choy in this study. Images of bok choy are collected and used as input data for training the neural network. The trained neural network is used for detecting bok choy and drawing bounding boxes around them, which generates bounding box coordinates and associate class probabilities as the output. In the second stage, a colour index-based segmentation is performed on vegetation (pixels) outside the bounding boxes using colour information, returning visual classification regarding the presence of weeds in the image. To extract weeds from the background, a colour index was determined and evaluated through GAs according to Bayesian classification error. In this way, the model focuses on identifying only the vegetables and thus avoiding handling various weed species.

[1] The first of the proposed systems in this paper is reconstruction of the 3D model of crops from the video of the crops and the second step is the detection and classification of weeds/crops. The videos are used for extraction of a number of frames using OpenCV. The desired number of frames containing the crops and weeds are obtained from the video by varying the number of frames extracted per second from the video. Manual tagging of weeds and crops is done on the extracted frames obtained from the video for creating the training dataset for the classifier. Manual tagging is done in segregation of crops and weeds, which is required for creating the training dataset. The tagged frames are processed for making the frames of the same size and resolution using OpenCV.

Pest Identification and Detection

The various steps involved in classification of insects were explored in this paper [8]. Image augmentation is applied to the insect dataset images to expand the training dataset, then the shape features are extracted from the insect images. Machine learning algorithms like ANN, CNN, SVM were applied to classify different insect classes. Classification of insects is performed based on the finite shape features extracted from the insect images. The insect images in the form of RGB converted to grayscale images for further feature extraction. Image processing techniques are applied to extract the shape features using the Sobel edge detection algorithm and morphological operations. The nine shape features include area, perimeter, major axis length, minor axis length, eccentricity, circularity, solidity, form factor, and compactness stored in feature vectors and then applied to the classifier models. The results showed that the CNN model provided the highest classification accuracy.

[13] This paper proposes a methodology for the prediction of pest insect appearance, which uses the environmental parameters (temperature and relative humidity) as input and machine learning for data processing and output generation. Depending on the environment various insects were caught, and observed. The approach in this paper predicts the appearance of insects in a period of three to five days when the user can react and protect the crops.

Species Recognition

Three methods are proposed in [22], namely Overall Mean-LBP, Region Mean-LBP, and ROM-LBP, based on the LBP for recognition of different types of plant leaves. The Overall Mean-LBP method extracted texture features from plant leaves using the R colour channel, the Region Mean-LBP method extracted texture features using the G colour channel. The ROM-LBP method is a combination of both the Region Mean-LBP and Overall Mean-LBP methods. These proposed methods calculated performances by using the ELM method. The parameters of this classifier were determined by an experimental approach using sigmoid as the activation function, and hidden layer neuron number in the range of with step size 100. The experimental results showed that the proposed methods have higher classification accuracy than the original LBP, improved LBP, and other descriptors for both noisy and noiseless images.

[16] This paper states that among different pre-trained CNN models, MobileNet, Xception and DenseNet-121 architectures are selected to form a bilateral CNN. The individual architectures (M1 and M2) in the bilateral CNN are used in the process of feature extraction. The models considered for M1 and M2 are MobileNet, Xception and DenseNet-121. The bilateral CNN is studied under two categories namely, Homogenous DL (same archi-tecture is used as M1 and M2, named as MoMoNet, XXNet, DeDeNet) and Heterogenous DL (different architecture is used as M1 and M2, named as MoXNet, XDeNet, MoDeNet). The proposed Bilateral Convolutional Neural Network performs the recognition of plant species.

[26] This paper primarily uses CNN for object detection. This algorithm takes in image as input, assigns importance to various aspects in the image to differentiate one from the other. The regional proposal network (RPN) is proposed by the faster region-based network (Faster R-CNN) to generate amounts of anchors in two stages, which are richer proposals to improve accuracy slightly. While Faster R-CNN has been a milestone of the two-stage detectors, these algorithms have some disadvantages in dealing with resources and time consumption for large datasets, which are not suitable for subsequent realistic application of object detection. representative methods such as YOLOv2 simplify object detection as a regression problem, which directly predicts the bounding box

and associate class probabilities without proposal generation. Classification and detection have been increasingly proposed, showing good performance on different crops. For a two-stage detection network, a sparse set of candidate object boxes are first generated, and then they are further classified and regressed. Then a fast region-based convolutional neural network (Fast R-CNN) is proposed to combine the target classification with bounding box regression to solve multi-task detection.

Crop Quality Management

Quick-Bird satellite imagery is used in this paper [24] for detecting powdery mildew and leaf rust in winter wheat, and high accuracy results were achieved with severe infections at late growth stages. Image selection and acquisition, prescription map creation, variable rate application, and performance evaluation are discussed so that the methodologies can be directly used or modified for crop diseases. Numerous classification techniques were evaluated to differentiate infested from non-infested areas in airborne imagery. Two unsupervised and six supervised classification classifiers were compared for identifying cotton root rot from airborne imagery. The evaluation results showed that although all eight methods were equally accurate, the two unsupervised methods were easy to use; these methods were therefore recommended for cotton root rot identification. Although both airborne and satellite images have been successfully used to detect and map many crop diseases, early detection is still a challenge. In most cases, by the time remote sensing imagery can reveal any disease symptoms, damage may have already been done to the crop.

[19] This paper discusses the methods to enhance crop productivity while sustaining the ways that promise greater social, economic, and environmental security. By using SLM crop productivity can be increased by the effective management of limiting resources, such as land, water, biodiversity, and the environment, while the overall sustainability of an ecosystem is enhanced. SLM is defined as the adoption of land-use systems that through appropriate management practices enable land users to maximize the economic and social benefits from the land while maintaining or enhancing the ecological support functions of the land resources.

Irrigation Optimization

[2] This paper proposes loT based smart irrigation system that has the capability of regulating soil moisture level as per requirement and users can remotely monitor, control, and collect data through the online website. When considering all weather prediction models tested in this research, the Sliding Window algorithm and HMM cannot provide an acceptable level of prediction accuracy for rainfall (in ml). However, by adding more weather parameters other than temperature and humidity, this accuracy could be improved. Through experimental results it was observed that Linear Regression model is the best model to predict temperature and humidity for next hour and Sliding Window model is the best model to predict soil moisture level for next hour.

Soil Modelling

[23] In this paper, various soil dielectric has been developed and divided into theoretical, semi-empirical and empirical models. From which semi-empirical model is very efficient from all the three models. On this basis it was found that the soil texture has a great influence on the CRP at low frequencies and developed empirical algorithms. Six samples collected were studied and found out that the sample properties ranged from 1.24 to 1.58 gcm3 for bulk density and 9.74 percent to 19.13 percent for clay density. To avoid soil volume changes in procedure, a special fixture was used to ensure that the high-temperature probe was in full contact with the soil. Experimental results show that the unfrozen water content can be different in the freezing process compared with that in the melting process at the same physical temperature. Several shortcomings in the studies could influence the performance of our improved model. This study did not directly measure the soil unfrozen water during the experiment; therefore, the parameterization scheme of unfrozen water content could be further optimized by using the NMR (nuclear magnetic resonance) technique, which is suggested for our future studies. Model predictions were then compared with dielectric measurements, showing a promising improvement. The correlation coefficients between measurements and the improved model predictions ranged from 0.7944 to 0.9865.

The approach proposed in this model for soil modelling is composed mainly of three types [25], Soil moisture, Vegetation, and Surface roughness. We have analysed the CYGNSS data and explored the correlation of the data with the collocated SMAP soil moisture and NDVI-derived climatology VWC. Through the comparative analysis of data and model, we have derived the model coefficients empirically by examining the dependence of the data on VWC and soil moisture. It can be concluded that the soil moisture dependence of the CYGNSS reflectivity agrees well with the characteristics of the Fresnel reflectivity computed with the Mironov dielectric constant model, which has been used by the SMAP and SMOS missions for soil moisture retrieval from brightness temperatures.

In this paper, some engineering models to estimate the soil permittivity are described and a new model intended for the UHF range is proposed [14]. It consists of messier model which estimates the frequency dependence of the permittivity and the conductivity of soils between 100 Hz and 1 Mhz, Topps Equation which relates the relative permittivity value between 0.1 and 1 GHz and the volumetric water content of soil , Roths Equation which provides the relationships between dielectric permittivity and moisture for mineral and organic soils for time domain reflectometry (TDR) applications, and Proposed model which calculates the complex permittivity of the soil. Soils with organic content higher than 10 percent were considered as organic soils. The proposed model, which is based on the well-known Messier and Debye models and Roths equations, provides a simple way to estimate the soil permittivity in the UHF from measurements of low-frequency conductivity and volumetric water content.

Soil Treatment Optimization

Soil structure is the most important parameter in grounding studies. The Wenner 4-Electrode Method is one of the widely used soil resistivity measurement techniques [21]. The soil resistivity measurements are performed by injecting current through two outer current electrodes, C1 and C2, and measuring the resulting voltage between two potential electrodes, P1 and P2. The focus of this paper is the treatment of the measured apparent soil resistivity data when the current and potential electrodes are far apart; specifically, from a given electrode spacing the methodologies present different soil structures where the ground grid will be installed. The methodology characterized by the arithmetic average of the measured apparent resistivity resulted in the reduction of the values when the electrodes spacing ranged from 2m to 8m.

[10] In this paper, we talk about the effects of salt content on measurement of soil resistivity. Ground resisting is a technique used to determine the effectiveness of a certain ground that will be used for grounding spot for buildings. A good grounding is not only for safety measures but also to prevent damage to industrial plants and equipment. For this research, the Earth Tester Type 3235 equipment was used to determine the resistivity of the soil. The Earth Tester Type 3235 consists of 3 electrodes which are the ground conductor, the bonding of the conductor to the ground electrode, and the ground electrode itself. Salt water treatment particularly NaCl is found to be an effective way to reduce the value of ground resistance but the withdrawal of this method is that it only gives temporary effects and can be unreliable especially for countries like Malaysia which has large numbers of rains throughout the year. This treatment showed that the higher density of NaCl will result with a lower resistivity of the soil. Therefore, when using this method to maintain low resistivity of a grounding, regular monitoring needs to be made to ensure the resistivity is at its lowest value. Furthermore, NaCl needs to be added regularly especially during the monsoon seasons.

Surface roughness has significant impacts on the land surface microwave emission and is an important process that should be considered in the development of soil moisture retrieval algorithm [12]. The surface roughness can be descripted by two parameters: rms height (h) and correlation length (l). A systemic method was developed to estimate surface roughness parameters at satellite footprint scale and apply it to retrieve surface soil moisture over the Tibetan Plateau from AMSR-E/AMSR2 observation. A regression relationship was developed between the optimized parameters and calibrated parameters and applied to correct the optimized parameter over the whole Tibetan Plateau.

Crop Yield Prediction

[17] In this paper, advanced Regression techniques of Lasso, Enet and Kernel

Ridge are used. They are further stacked to minimize the error and obtain better results. Each based model is trained on 4 folds in each iteration and the holdout fold is used for prediction. The data of more than a hundred of crops planted around the world country has been acquired from the Indian Government Repository. The attributes in the data are State, District, Crop, Season, Year, Area and Production with around 2.5 lakh observations. Use of Stacked regression has been improvised more than when those models were applied individually. It creates a robust model by combining the predictions of multiple models.

SVM classification is done based on the texture, shape, colour of patterns on the diseased surface as it includes an unambiguous perception of the defects [18]. DNN for CYP for determining an accurate yield prediction model. The model performed fundamental understanding for setting up the relation among the yield and the interactive factors with respect to the powerful and comprehensive algorithm. The results showed and suggested that the regression trees outperformed better when compared with existing supervised models. The developed model (CNN, MCNN) reduced the relative error as well as decreased the prediction efficiency of crop yield.

Table 2.1: Weed Detection Literature Review

Algorithm	Advantages	Disadvantages	Datasets	Performance metrics
WIS	Low cost processing with high accuracy.	Using only RGB images to achieve crop and weeds classification is prone to error.	Farmland of about 2,100sq mt in Luye, Taitung, Taiwan,	Accuracy: 98.8%
LBP+CNN	Robust to monotonic gray-scale changes caused.	It produces long histograms, which slow down the recognition speed especially on large-scale face database	Farmland of about 2,100sq mt in Luye, Taitung, Taiwan,	Accuracy: 96%
CNN	Good at obtaining high accuracy image features.	Overfitting, exploding gradient, and class imbalance	Farmland of about 2,100sq mt in Luye, Taitung, Taiwan,	Accuracy: 92%
HOG	Not mentioned	No efficiency in weed detection	Farmland of about 2,100sq mt in Luye, Taitung, Taiwan,	Accuracy: 85%
Resnet50	Tackles the vanishing gradient problem using identity mapping.	Slow processing	Cucumber dataset, Onion dataset	Accuracy: 88.8%, Accuracy: 90%(Onion dataset), Accuracy: 84.6%(Cucumber dataset)

Table 2.2: Crop Yield Prediction Literature Review

Algorithm	Advantage	Limitation	Dataset	Performance Metrics
SVM	Cascade of two SVM classifiers for achieving the accuracy, specificity and precision metrics	The developed model was not given the proper extensive analysis of the defective outlines.	[7]	Accuracy = 97.77% Sensitivity = 96.55% Precision =99.24%
CNN, MCNN	Utilizes spatial features as input and trained by backpropagation to reduce error of prediction.	Reduces the relative error as well as decreases the prediction efficiency of crop yield.	[7]	MCNN RMSE value = 1396.4 Relative Error=9.8465
H-ANN,	Not Mentioned	It is incapable of capturing the nonlinear bond between input and output variables.	[7]	RMSE=4.72
ANN and MLR.	It uses a combination of backpropagation algorithms with ANN to evaluate the crop yield with higher precision.	The developed model showed difficulties in training the neural network model	[7]	MLR RMSE=9.8% MAE=6.9% R=89% ANN RMSE=5.1% MAE=6.4% R=99%

Table 2.3: Irrigation Optimization Literature Review

Algorithm	Advantages	Disadvantages	Datasets	Performance metrics
Sliding Window	Not Mentioned.	Cannot provide an acceptable level of prediction accuracy for rainfall.	Historical weather data set sensor weather data set	Accuracy: 85%(method 1) Accuracy: 71%(Method 2)
HMM	Calculates probability of weather not one day ahead but beyond that.	Cannot provide an acceptable level of prediction accuracy for rainfall.	Historical weather data set sensor weather data set	Accuracy: 75.5% (sub-model 2) Accuracy: 89.9%
Linear Regression	Easier to implement, efficient to train	It is often quite prone to noise and overfitting	Historical weather data set sensor weather data set	Prediction error: 0.5(temp) Prediction error:1.5 (humidity)

Table 2.4: Disease Detection Literature Review

Algorithm	Advantages	Disadvantages	Datasets	Performance metrics
CNN LeNet architecture	Extensively uses CNN designs to increase accuracy	Overfitting in some cases and no built-in mechanism	PlantVillage (extended) No. of Classes:3	Accuracy: 92â€“99%
CNN AlexNet, GoogLeNet	GoogleNet has 22 layers which increases the accuracy	Requires more time to train.	PlantVillage (extended) No. of Classes: 9	Accuracy: 99%
CNN ResNet34	Large number of layers can be trained without increasing the training error percentage.	Requires more time to train the model.	PlantVillage. No. of Classes: 10	Accuracy: 96.4%
CNN GoogleNet	It has a top-5 error rate of 6.67% in classification tasks.	Not Mentioned	Field. No. of Classes: 5	Accuracy: 89.5%
AlexNet, GoogLeNet, Inception-v3, ResNet-50	Has deeper architecture with 8 layers which means it is better able to extract features, and faster training of models.	Not Mentioned	Self-captured and PlantVillage datasets. No. of Classes : 5	Accuracy: 98%, 96%, 98%, 99%
CNN AlexNet	Recognized off-center objects with precision.	Requires more processing power to train the model.	Rice field images No. of Classes: 10	Accuracy: 95%

Table 2.5: Pest Prediction and Detection Literature Review

Algorithm	Advantages	Disadvantages	Datasets	Performance metrics
LSTM	Provides a large range of parameters Hence, no need for fine adjustments.	Requires more memory to train.	Climate Data for pest prediction	Accuracy: 89%
ANN	Supports automatic identification, and classification	It does not guarantee convergence on a prediction or solution.	The Wang and Xie datasets	Accuracy:71%
CNN	Useful to recognize the insects according to classes and family.	Required more computation time for classification	The Wang and Xie datasets	Higher accuracy is achieved 91.5%-93.9%
SVM	Effective in minimizing the overall interspecific error rates in classification	Doesn't perform well on large datasets because the required training time is higher.	The Wang and Xie datasets	Accuracy: 80%
KNN	Does not use any parameters and achieves a better identification rate for certain insects	Sensitive to noisy and missing data.	The Wang and Xie datasets	Accuracy: 78%
Naive Bayes	Provides better accuracy for multi-class prediction problems.	Assumes all features are independent or unrelated, so it cannot learn the relationship between features.	The Wang and Xie datasets	Accuracy: 54%
QDA	Performs better with limited number of training observations	It is not as flexible as KNN	Climate Dataset	Accuracy: 79%
Decision Tree	It requires less data preprocessing.	It has High variance.	Climate Dataset	Accuracy: 86%
Random Forest	It runs efficiently on a large dataset.	It is found to be biased while dealing with categorical variables.	Climate Dataset	Accuracy: 82.7%
AdaBoost	Combines weak classifiers to improve the accuracy.	It needs a quality dataset	Climate Dataset	Accuracy: 87.1%

Chapter 3

System Analysis

3.1 Functional Requirements:

Proposed system comprised of following Functional Modules

- Pre-Processing and data acquisition:
  1. Collect Images and Data:
     The system will gather images as well as numerical data for analysis.
- Predict Crop Yield:
  1. Read all required parameters like (ph level of the soil, nitrogen levels, potassium levels, and phosphorous levels in the soil).
  2. Process input information and generate prediction results.
- Species Recognition:
  1. Read input images and preprocess
  2. Extract Features and generate classification results.
- Weed Detection and Prediction:
  1. Read input images and preprocess the images.
  2. Extract Features and classify crops as weed or not.
- Disease Detection and Identification:
  1. Read input images and preprocess the images.
  2. Extract Features and classify crops as weed or not.
- Pest Detection and Identification:
  1. Read input images and preprocess the images.
  2. Extract Features and Detect or identify the pests.

- Crop Recommendation

    1. Read all required parameters like (ph level of the soil, temperature, humidity, nitrogen, potassium and phosphorus levels in the soil).
    2. Process input information and generate prediction results.

3.2 Non-Functional Requirements:

These are basically the quality constraints that the system must satisfy according to the project contract.The priority or extent to which these factors are implemented varies from one project to another. They are also called non-behavioral requirements.

- Maintainability: The system should go under regular maintenance to ensure that the system works correctly.
- Availability: The system should be available at all times.
- Accuracy: The system should provide better accuracy.
- User friendly: The system should be easier to navigate through.
- Scalable: The system should be scalable with a scope to add various features in future.

3.3 Specific Requirements

Hardware

The hardware environment consists of the following:

- Processor: Minimum 1 GHz; Recommended 2GHz or more.
- Ethernet connection (LAN) OR a wireless adapter (Wi-Fi)
- Disk Space: Minimum 2 GB; Recommended 4 GB or more Memory
- RAM: 4 GB RAM minimum , 8 GB RAM recommended
- Esp32 / DIR/ Raspberry Pi camera

Software

The software environment consists of the following:

- Python, Machine Learning, Deep Learning, Flask, Bootstrap5
- Database: PostgreSQL

3.4 Use Case Diagrams and Description

Use Case Diagram:

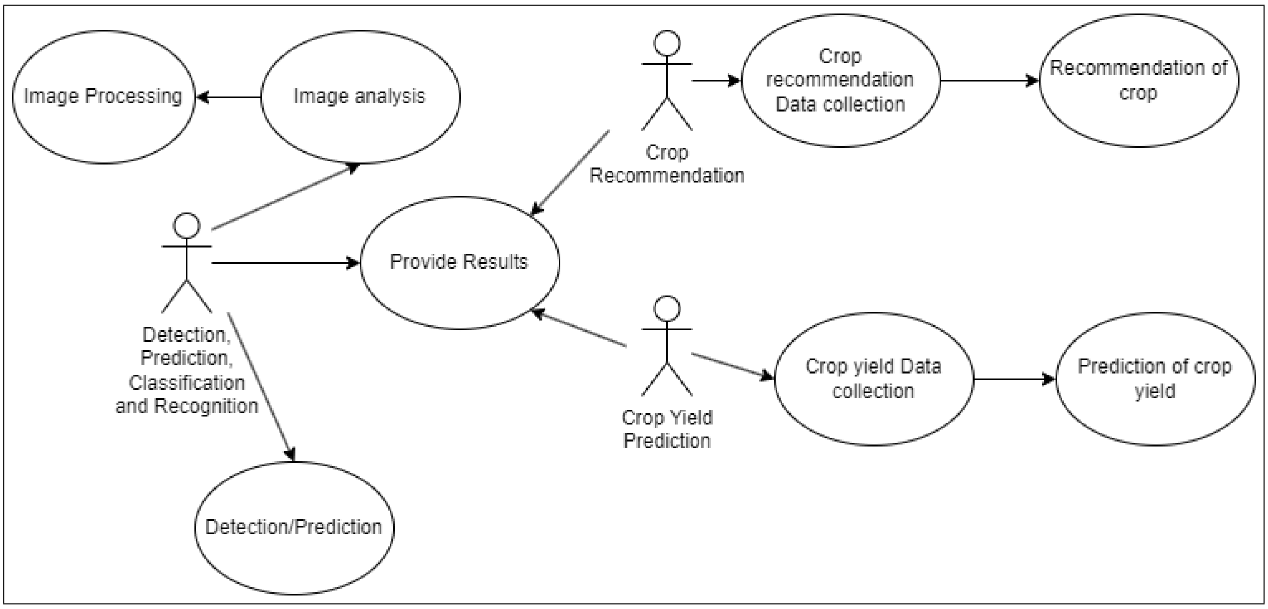


Figure 3.1: Use Case Diagram of Precision Agriculture

Description: As shown in the Figure 3.1, the user mainly performs three tasks as shown by each actor. The system takes the image as input and performs pre-processing, followed by classification or detection. Following Tables from 3.1 to 3.8 are presenting each use case and its description

Table 3.1: Use case 1- Image Processing

Use Case	Image Processing
Use Case ID	UC01
Actor	Detection, Prediction, Classification and Recognition
Descriptor	Processing of images happens at this stage.

Table 3.2: Use case 2- Image Analysis

Use Case	Image Analysis
Use Case ID	UC02
Actor	Detection, Prediction, Classification and Recognition
Descriptor	The system analyzes the input image.

Table 3.3: Use case 3- Detection/Prediction

Use Case	Detection/Prediction
Use Case ID	UC03
Actor	Detection, Prediction, Classification and Recognition
Descriptor	The model takes image input and provides detection and prediction results.

Table 3.4: Use case 4- Provide Visualization

Use Case	Provide Visualization
Use Case ID	UC04
Actor	All
Descriptor	A dashboard is made available which displays various results in pictorial format.

Table 3.5: Use case 5-Crop recommendation data collection

Use Case	Crop recommendation data collection
Use Case ID	UC05
Actor	Crop Recommendation
Descriptor	The model takes the crop recommendation dataset and provides crop recommendations.

Table 3.6: Use case 6- Recommendation of crop

Use Case	Recommendation of crop
Use Case ID	UC06
Actor	Crop Recommendation
Descriptor	Suggest suitable crops based on the dataset.

Table 3.7: Use case 7- Crop Yield Data Collection

Use Case	Crop yield data collection
Use Case ID	UC07
Actor	Crop Yield Prediction
Descriptor	The model takes the crop yield dataset and predicts future crop yield.

Table 3.8: Use case 8- Prediction of crop yield

Use Case	Prediction of crop yield
Use Case ID	UC08
Actor	Crop Yield Prediction
Descriptor	Predict yield of crops based on dataset.

Chapter 4

Analysis Modeling

4.1 Flow Diagram

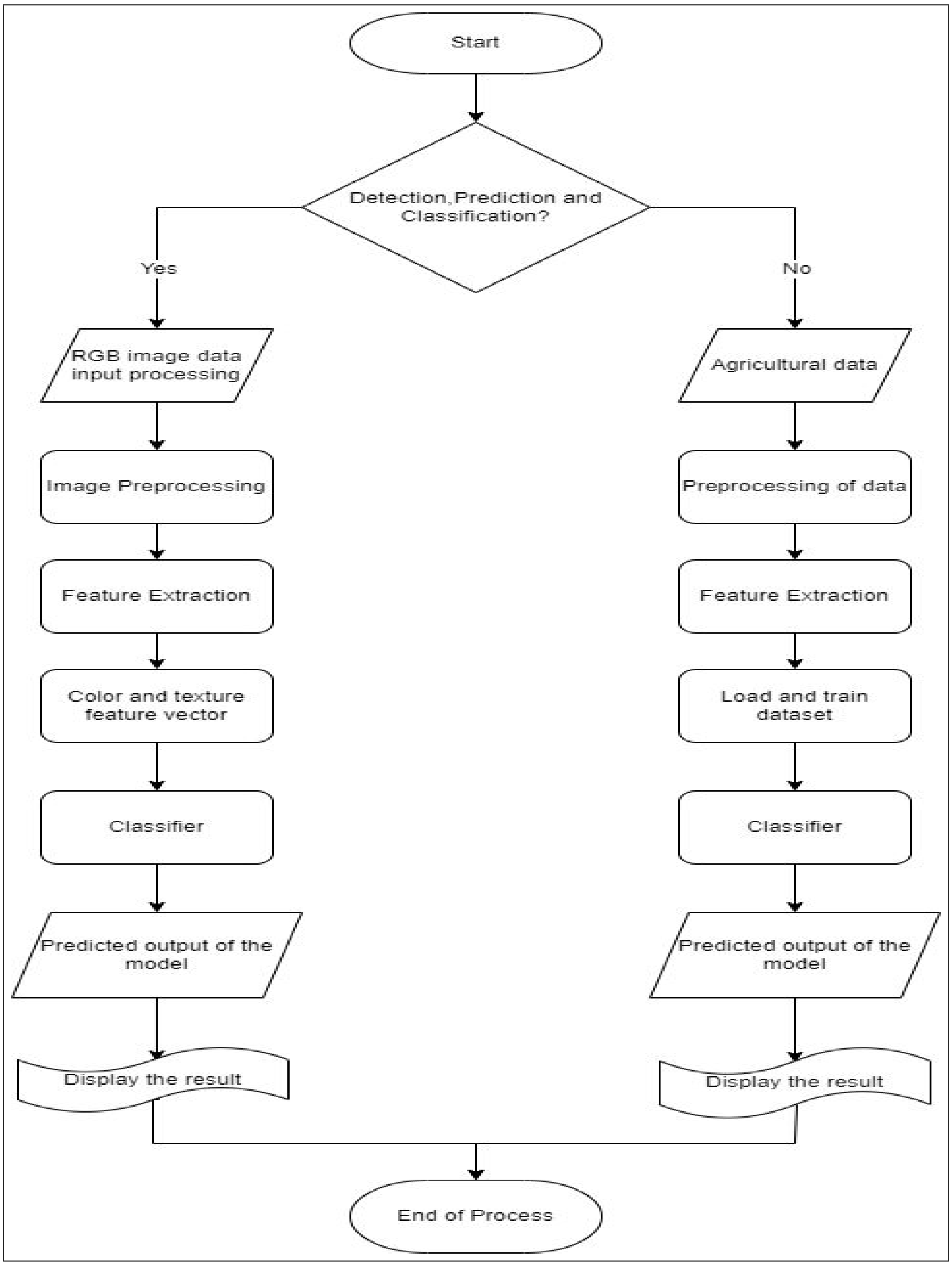


Figure 4.1: Precision Agriculture Workflow

Description: As shown in the Figure 4.1, this flowchart defines the flow of the process that takes place in the procedure of Precision Agriculture.

4.2 Activity Diagram:

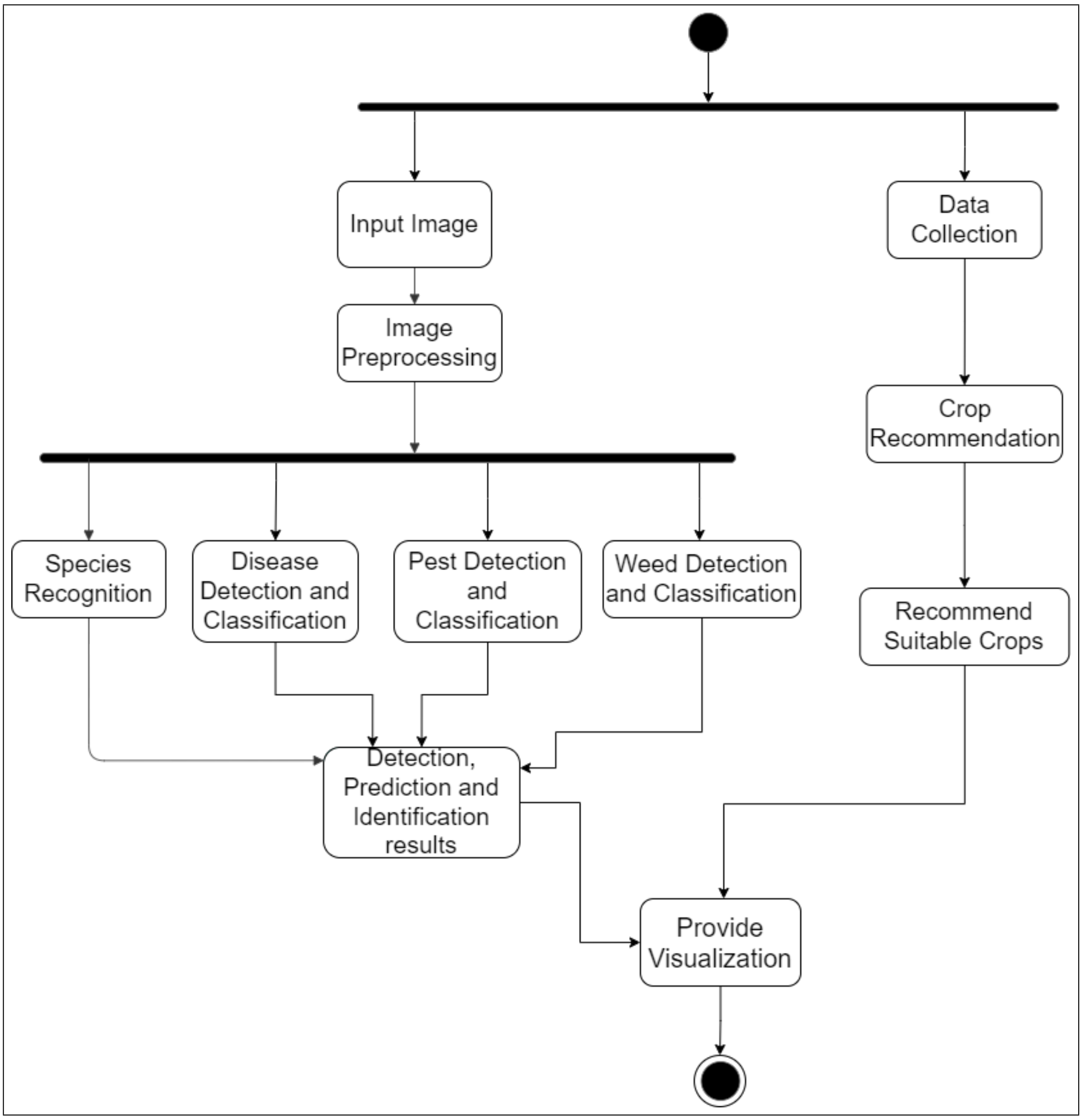


Figure 4.2: Activity Diagram of Precision Agriculture

Description: As shown in the Figure 4.2, login/registration is done initially then the modules separate based on the type of input data they take: image input and numeric data. Then the algorithms are segregated based on that with their functions.

4.3 Data Flow Diagrams

4.3.1 DFD: Level 0

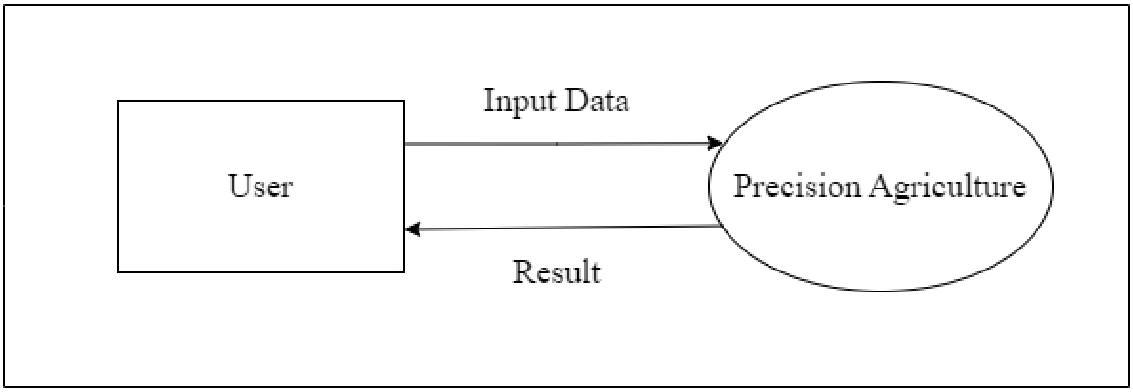


Figure 4.3: DFD Level 0

Description: As shown in Figure 4.3, the user uploads the images. The system then processes the images to display the result to the end user.

4.3.2 DFD: Level 1

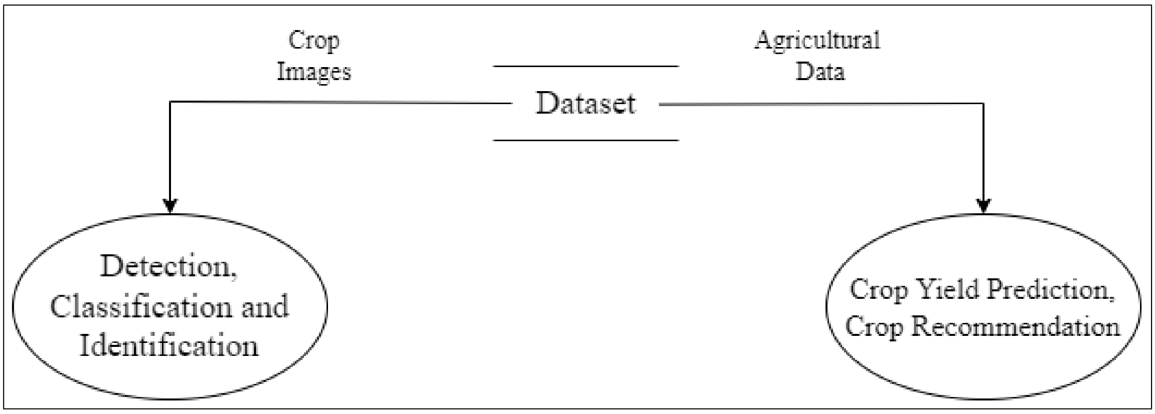


Figure 4.4: DFD Level 1

Description: As shown in Figure 4.4, the data collected is segregated into crop images, sensor data, and agriculture data used for Detection/classification/identification, Crop quality/irrigation management, and Crop yield prediction respectively.

4.3.3 DFD: Level 2

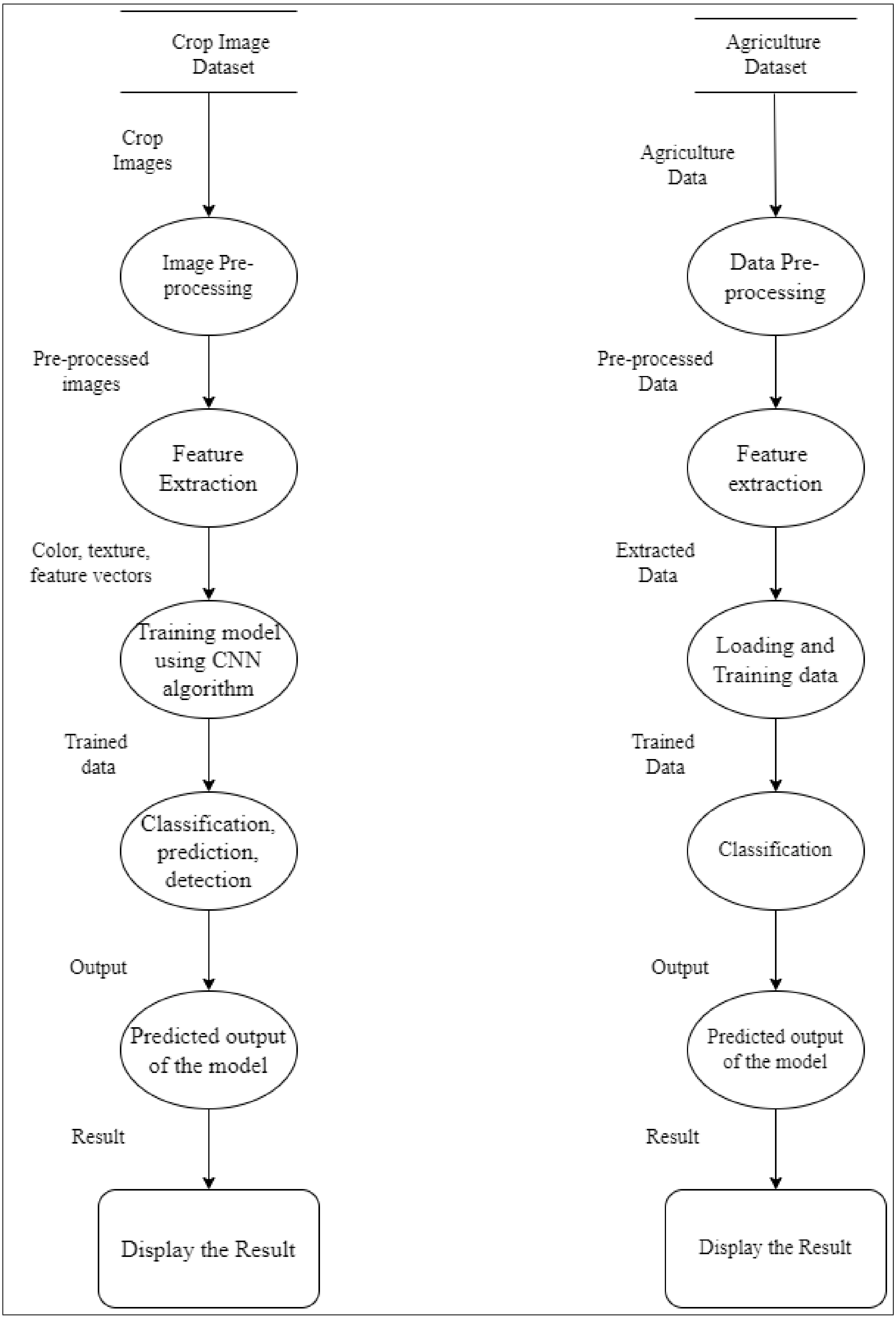


Figure 4.5: DFD Level 2

Description: As shown in Figure 4.5, the crop images, sensor data, and agriculture data are preprocessed respectively, then further for crop images and agriculture dataset features are extracted, and the dataset is trained using CNN and SVM algorithms respectively. Once the classification/detection is done the predicted output is then returned to the user. For sensor data the data is first organized, then mapped and integrated for training for classification. Once classified the results are displayed to the user.

4.4 TimeLine Chart

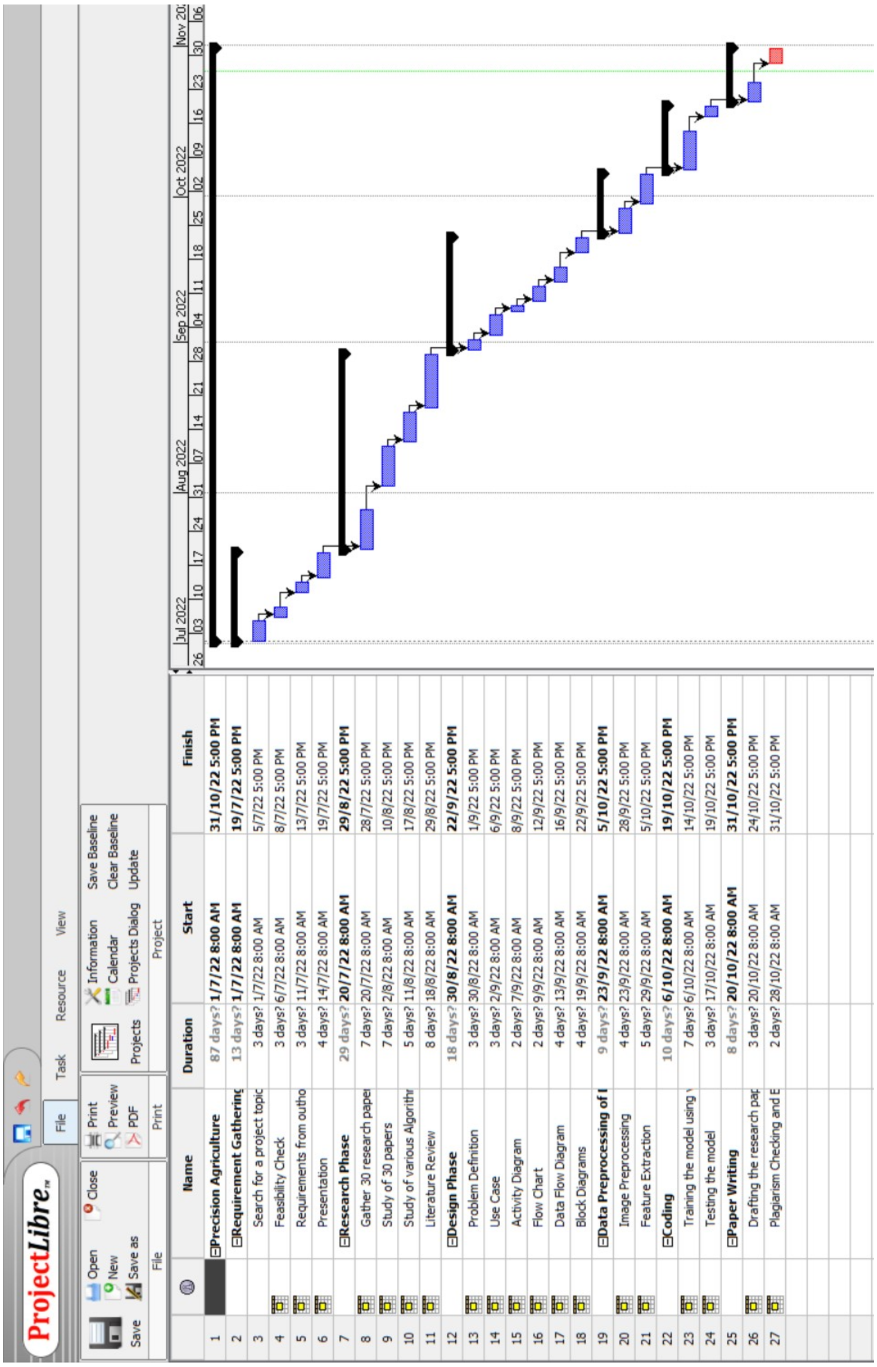


Figure 4.6: TimeLine chart of odd semester

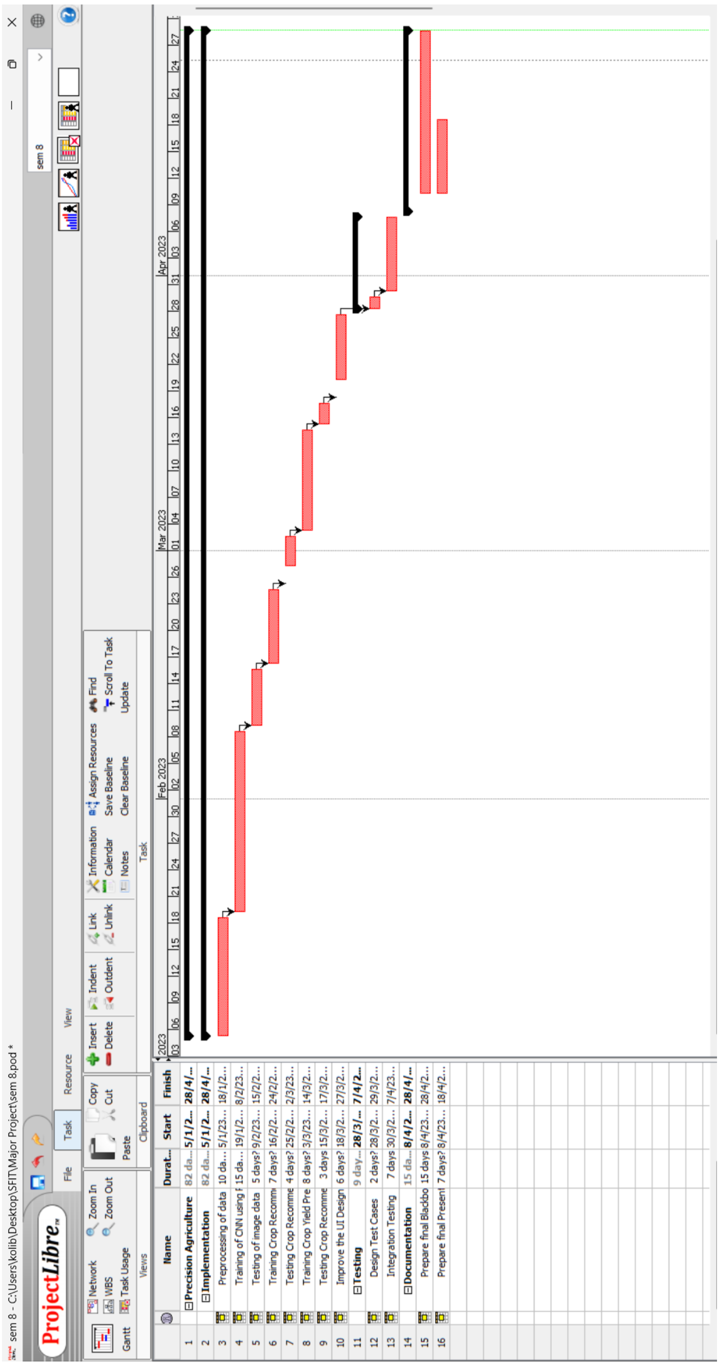


Figure 4.7: TimeLine chart of even semester

Description: The Figure 4.6 and 4.7 is the TimeLine Chart depicting the different tasks that were performed and the resources used over time, which helps to plan in an efficient way, coordinate, and track some particular tasks in the project. We have used the Gantt chart to illustrate project tasks and events in a chronological order, it shows which tasks were performed by whom and when. The series of horizontal lines present show the amount of work done or production completed in a given period of time in relation to the amount planned for those projects.

Chapter 5

Design

5.1 Architectural Design

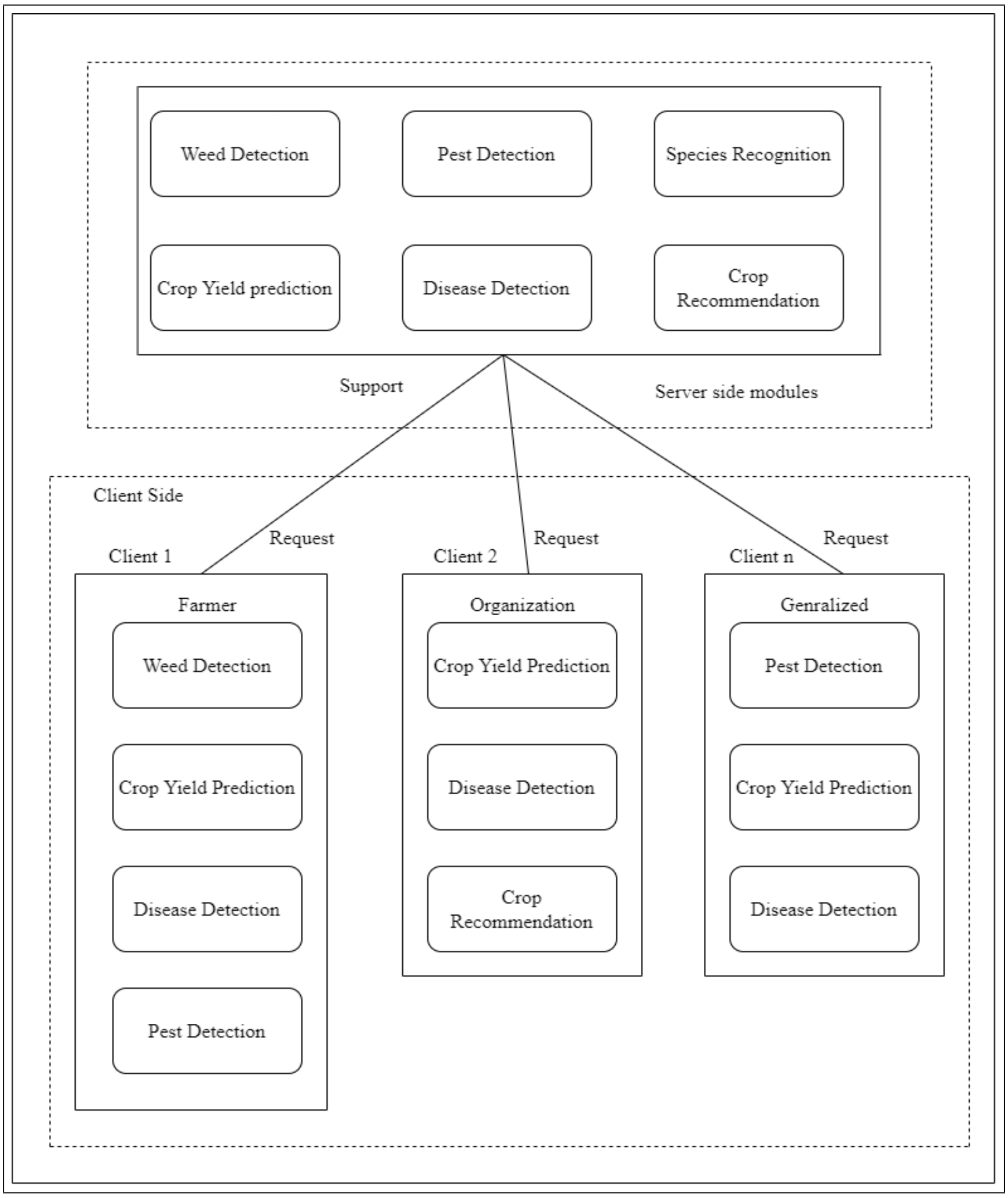


Figure 5.1: Architecture Diagram of Precision Agriculture

Description: As shown in Figure 5.1,the above figure represents the Client - Server architecture of the system . The Clients include Farmers and Organizations. The Server side provides the various modules to the Clients.

5.2 Block Diagram

Various modules are integrated into precision agriculture system. Following section presents the working principle along with the various phases in each of these modules.

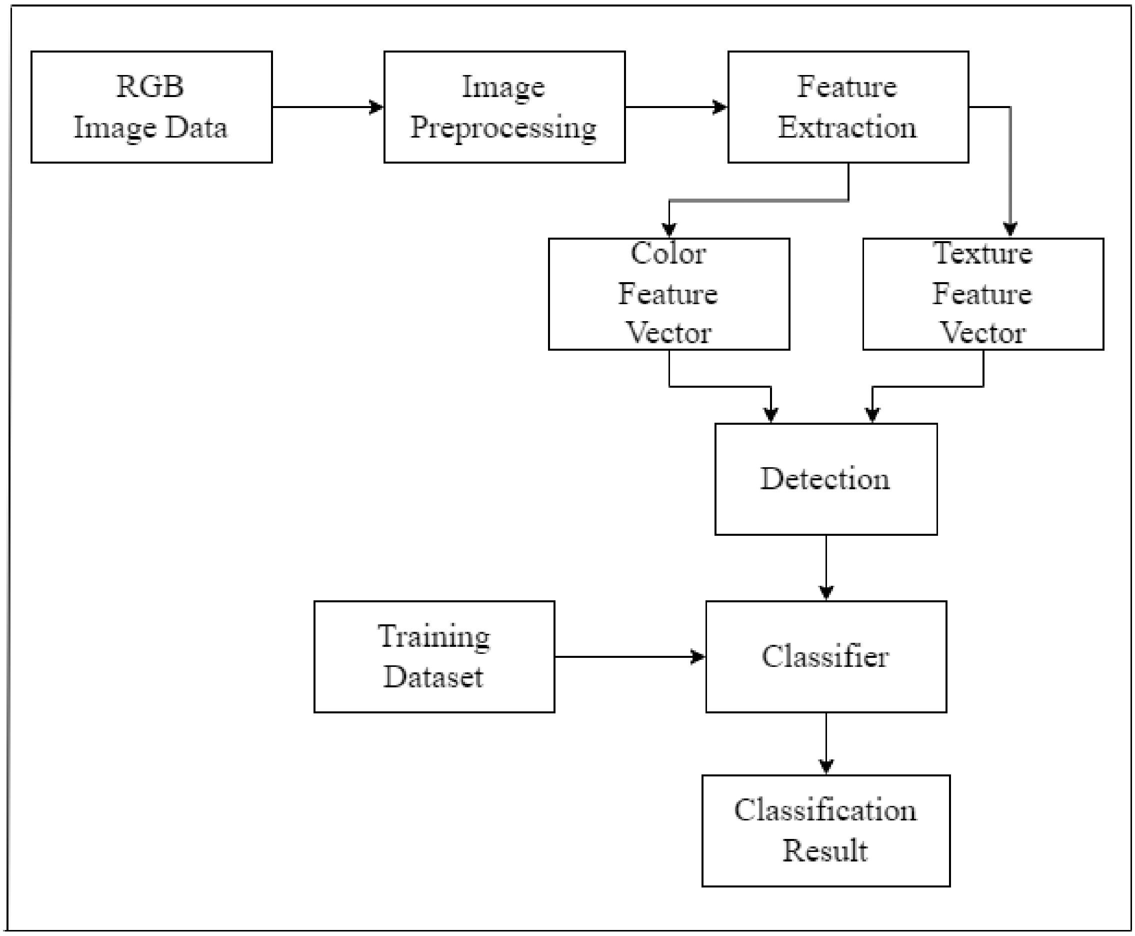


Figure 5.2: Block Diagram of Classification and Detection Modules

Description: As shown in the Figure 5.2 the process of classification and detection modules. The RGB data is taken and then image preprocessing is performed. Then feature extraction is done which gives us two features: color feature vector and texture feature vector. Then a classifier is used to get the results.

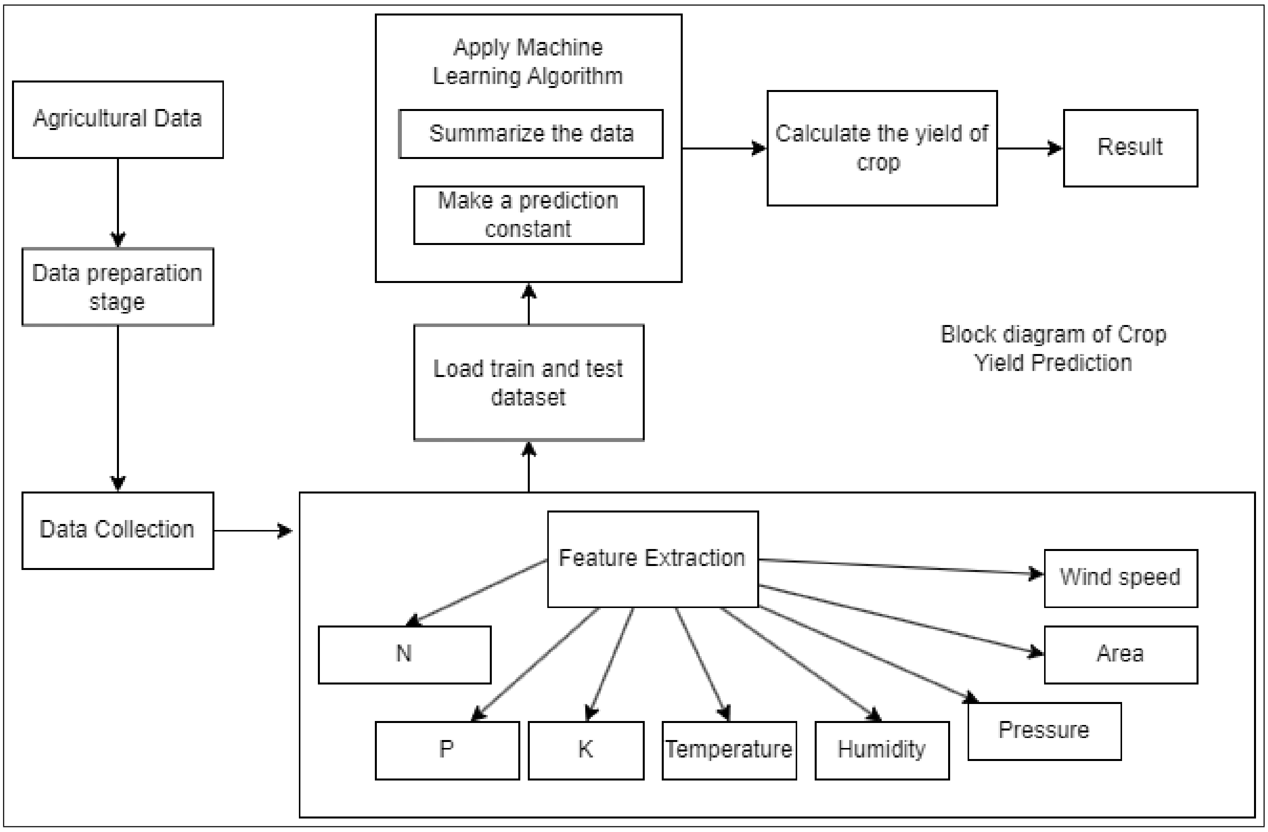


Figure 5.3: Block diagram of Crop Yield Prediction

Description: As shown in the Figure 5.3 the process of the Crop Yield Prediction module. The sensor data is taken and then organization of data is performed. Then data mapping and integration is done. Then a classifier is used that is SVM in our case and parameters are modified if the output is not feasible. And finally the crop yield result is displayed.

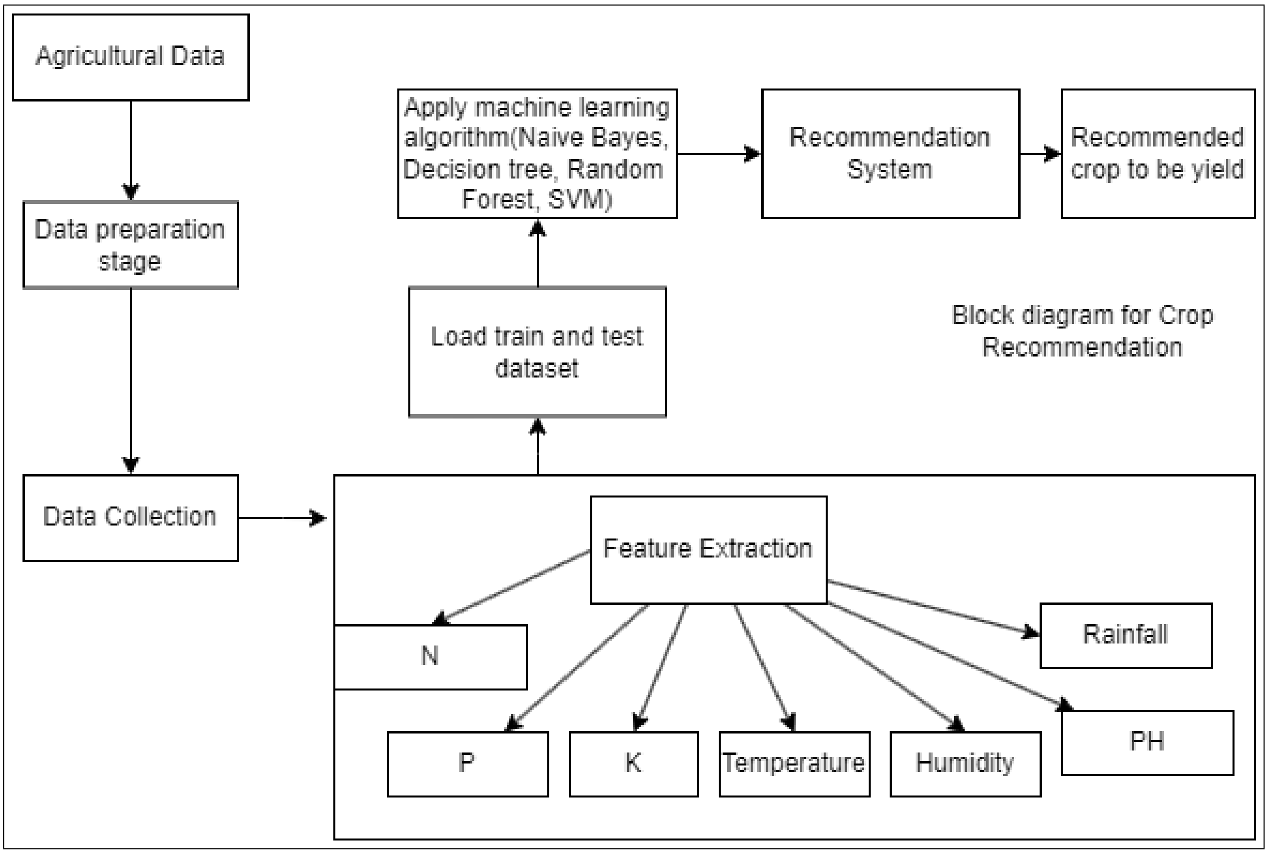


Figure 5.4: Block diagram of Crop Recommendation

Description: As shown in the Figure 5.4 the process of the Crop Recommendation Management module. The sensor data is taken and then organization of data is performed. Then data mapping and integration is done. Then a classifier is used that is Random Forest in our case and parameters are modified if the output is not feasible. And finally the predicted values are displayed.

5.3 Algorithms Used

5.3.1 CNN Resnet50:

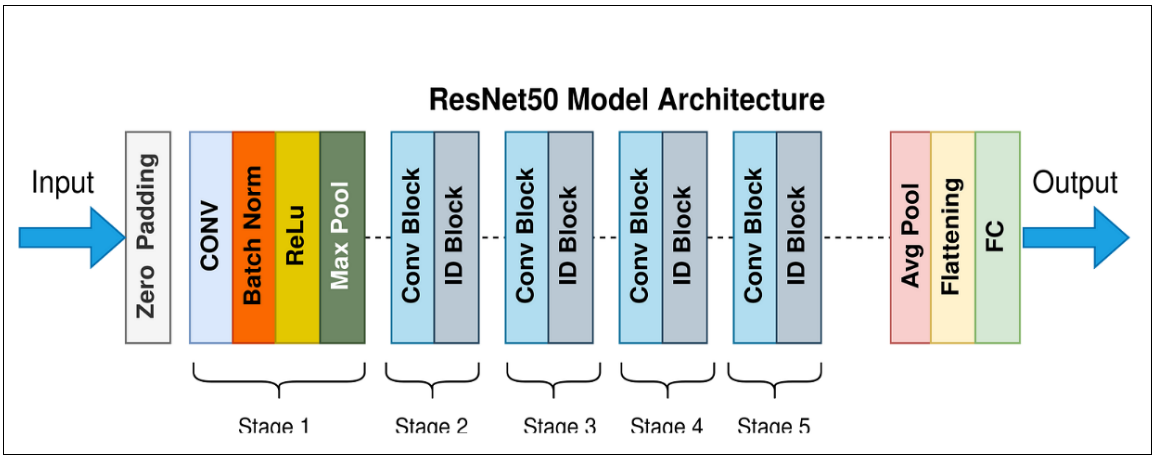


Figure 5.5: CNN Resnet50 Model [5]

As Shown in the Figure 5.5 CNN ResNet, or Residual Neural Network, is a deep neural network architecture that is specifically designed to overcome the problem of vanishing gradients in very deep neural networks. The idea behind ResNet is to use residual connections or skip connections, which allow the gradient to be directly propagated to earlier layers, making it easier to train very deep networks.

Here is an overview of the ResNet algorithm:

1. Input image is passed through a convolutional layer, followed by a batch normalization layer and a ReLU activation function.

2. This is followed by a series of residual blocks, each consisting of multiple convolutional layers, batch normalization layers, and ReLU activation functions.

3. The output of each residual block is added to the input of the block, allowing the gradient to be directly propagated to earlier layers.

4. The final output of the last residual block is passed through a global average pooling layer and a fully connected layer to produce the final classification output.

5.3.2 CNN DenseNet:

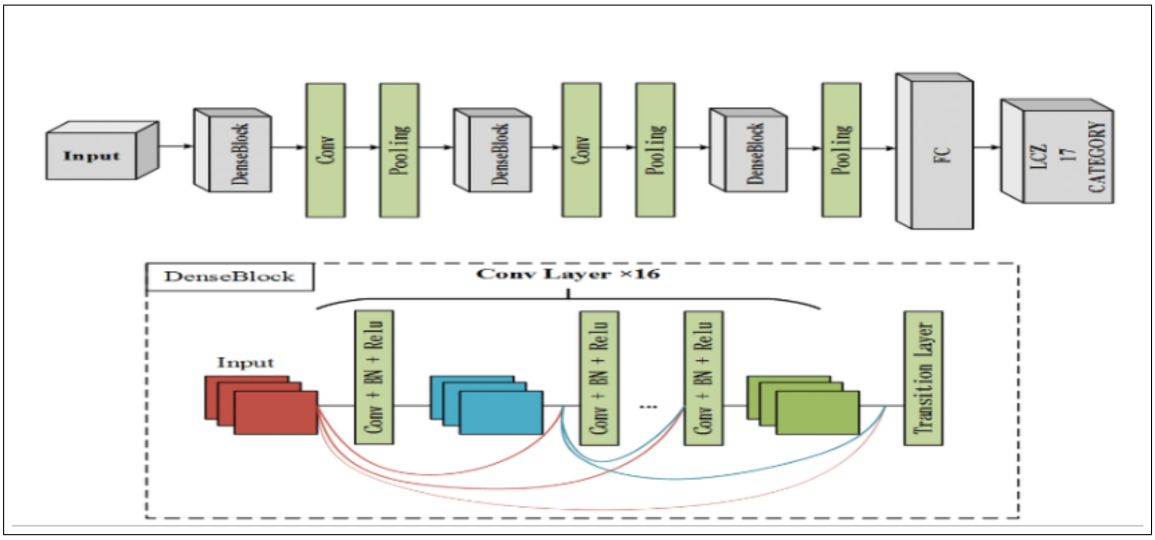


Figure 5.6: CNN DenseNet Model [4]

Figure 5.6 represents the DenseNet Model. The DenseNet architecture consists of multiple dense blocks, which are composed of multiple layers. Each dense block takes the output feature maps of the previous dense block as input and concatenates its own output feature maps with the input feature maps of the next dense block. This process of dense connectivity allows for efficient feature reuse and improves the flow of gradients, which in turn leads to better convergence and regularization.

Here is an overview of the DenseNet algorithm:

1. The input image is passed through a convolutional layer, followed by batch normalization and ReLU activation.

2. The output of the first convolutional layer is passed through a series of dense blocks, each of which consists of multiple convolutional layers, batch normalization, and ReLU activation.

3. The output of each dense block is concatenated with the input feature maps of the next dense block.

4. The output of the last dense block is passed through a global average pooling layer, which averages the feature maps over the spatial dimensions.

5. The output of the global average pooling layer is passed through a fully connected layer, followed by softmax activation to produce the final class probabilities.

6. During training, the network is optimized using backpropagation and stochastic gradient descent to minimize the cross-entropy loss between the predicted class probabilities and the ground truth labels.

5.3.3 Bins Approach:

Bins promising approach to retrieving crucial features from large digital image databases, by analyzing certain contents such as color, shape, texture and their representation as feature vectors in various other formats. The implementation of the algorithm has been explained in Chapter 6.2.

Bins Formation Process: 8 Bins [9]

Step 1: Spilt the image into R, G and B planes.

Step 2: Obtain the histogram for each plane

Step 3: Divide each histogram into 2 parts and assign a unique flag to each part (either "0" OR "1").

Step 4: To extract the color feature of the image, pick up the original image pixel and check its R, G and B values find out in the histogram that in which range these values exactly falls, and based on it assign the unique flags to the r, g and b values of that pixel with respect to the partition of the histogram it belongs.

Step 5: Count of pixels in the bin: Based on the flags assigned to each pixel with respect to the R, G B values and 2 partitions (e. g. 0 and 1) of the histogram we can have 8 combinations from **000 to 111** which are the **total 8 bins** following and implementing above algorithm we get 8 bins with the count of Pixels in each of the 8 bins with respect to the image under feature extraction process.

5.3.4 Classifiers Used on Bins Features :

The following classifiers have been used on the features extracted using the Bins Approach . Crop Yield Prediction and Crop Recommendation modules also use these modules to predict and recommend using the numerical agriculture data.

SVM(Support Vector Machine):

Figure 5.8 represents the SVM Architecture. SVM are used for classification, regression, prediction and densities of data,although in this case it was used for classifcation.

1. In this algorithm, we plot each data item as a point in n-dimensional space (where n is the number of features you have) with the value of each feature being the value of a particular coordinate.

2. Then, we perform classification by finding the hyperplane that will differentiate the two classes very well.

3. An ideal SVM analysis should produce a hyperplane that entirely separates the vectors (cases) into two non-overlapping classes.

4. To separate the two classes of data points, many possible hyperplanes could be chosen.

5. Our objective is to find a plane that has the maximum margin, i.e., the maximum distance between data points of both classes

Decision Tree :

Decision Trees are a popular machine learning algorithm that can be used for both classification and regression tasks. They work by recursively partitioning the input data into smaller and smaller subsets based on the values of the input features, until a stopping criterion is met.

Here is a high-level overview of the Decision Tree algorithm:

1. Input data is collected and preprocessed. The input data consists of one or more predictor variables and a target variable.

2. The Decision Tree algorithm starts by selecting the most important feature from the input data and creating a root node.

3. The data is then split into smaller subsets based on the values of the selected feature. Each subset is associated with a branch that emanates from the root node.

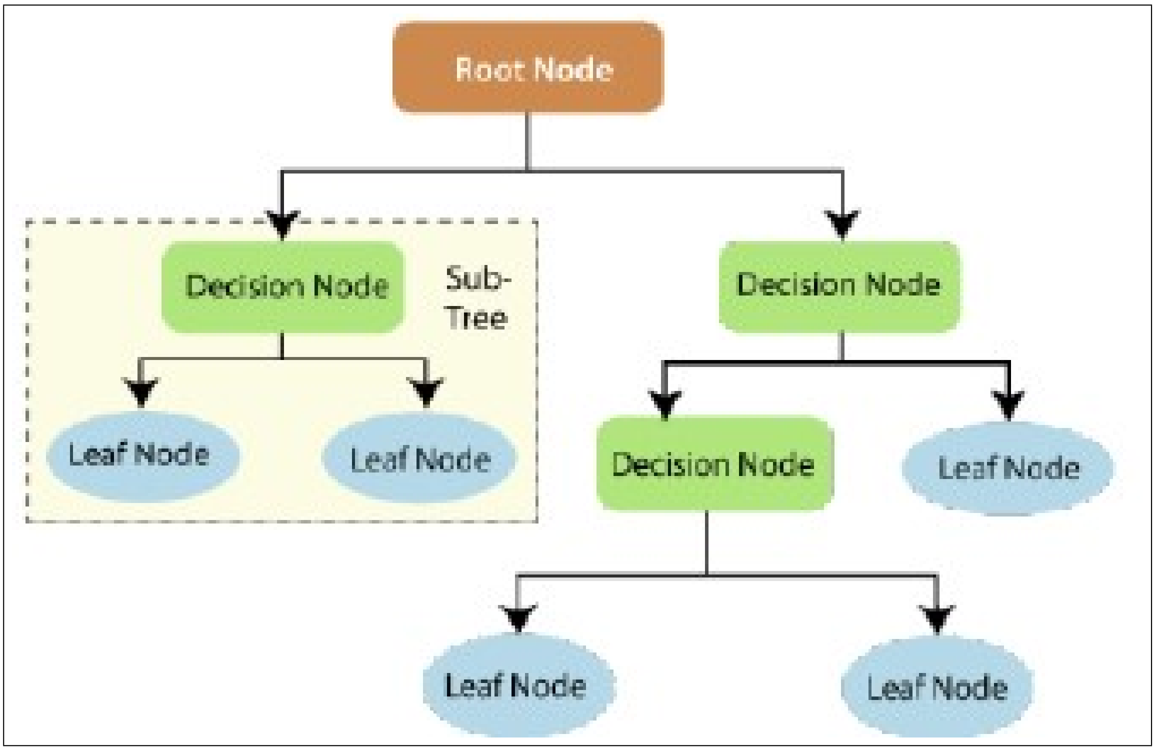


Figure 5.7: Decision Tree Classifier [3]

4. The algorithm continues recursively by selecting the most important feature from each subset and creating a new node for each subset.

5. The algorithm stops when a stopping criterion is met, such as when all the data in a subset belongs to the same class or when the depth of the tree exceeds a certain threshold.

6. Once the Decision Tree is built, it can be used to make predictions on new data by traversing the tree from the root node to a leaf node and using the class associated with the leaf node as the prediction.

Logistic Regression:

Logistic Regression is a widely used machine learning algorithm for classification problems. It is a supervised learning algorithm that is used to predict the probability of a binary outcome based on one or more predictor variables.

Here is a high-level overview of the Logistic Regression algorithm:

1. Input data is collected and preprocessed. The input data consists of one or more predictor variables and a binary outcome variable.

2. The logistic regression model is trained using the input data. The goal is to find the best set of parameters that maximize the likelihood of the observed data.

3. During training, the logistic regression model uses a sigmoid activation function to map the input data to a probability value between 0 and 1.

4. Once the model is trained, it is used to make predictions on new data by applying the learned parameters to the new input data and applying the sigmoid function to obtain the probability of the binary outcome.

5. The output of the logistic regression model can be thresholded to obtain a binary classification output.

Naive Bayes Classifier:

Naive Bayes Classifier is a popular machine learning algorithm for classification problems. It is based on the Bayes' theorem and assumes that the features are conditionally independent given the class variable.

Here is a high-level overview of the Naive Bayes Classifier algorithm:

1. Input data is collected and preprocessed. The input data consists of one or more predictor variables and a target variable.

2. The Naive Bayes Classifier algorithm starts by calculating the prior probability of each class based on the training data.

3. For each feature in the input data, the algorithm calculates the likelihood of the feature given each class.

4. The algorithm combines the prior probability and the likelihood of each feature to calculate the posterior probability of each class given the input data.

5. The class with the highest posterior probability is selected as the prediction.

6. The algorithm can be updated with new data by updating the prior probabilities and the likelihoods of the features.

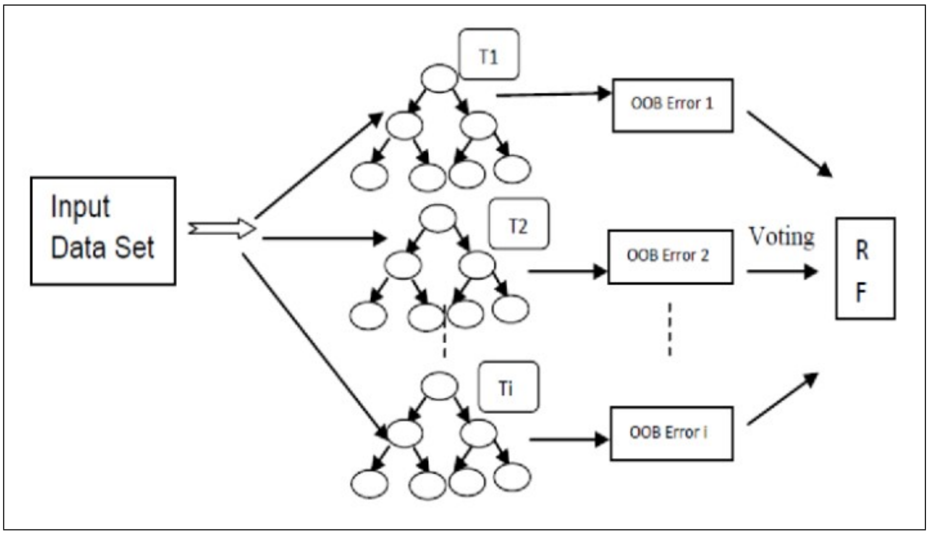


Figure 5.8: Random Forest Classifier[15]

Random Forest Classifier:

Random Forest is an ensemble machine learning algorithm that uses multiple decision trees to make predictions. The algorithm combines the predictions of multiple decision trees to improve the accuracy and reduce the overfitting. Here is a high-level overview of the Random Forest Classifier algorithm:

1. Create bootstrap samples: The algorithm randomly selects a subset of the data with replacement. This process is repeated multiple times to create bootstrap samples.

2. Create decision trees: For each bootstrap sample, the algorithm creates a decision tree. The decision tree is created using a random subset of the features. This helps to prevent the trees from becoming too similar to each other.

3. Grow trees: The decision trees are grown by recursively splitting the data based on the feature with the highest information gain or Gini index.

4. Predict the class: Once the trees are built, they can be used to make predictions on new data. The algorithm predicts the class by taking the majority vote of the predictions of all the trees.

5. Evaluate the performance: The algorithm evaluates the performance of the random forest using a validation set or cross-validation.

5.4 User Interface Design

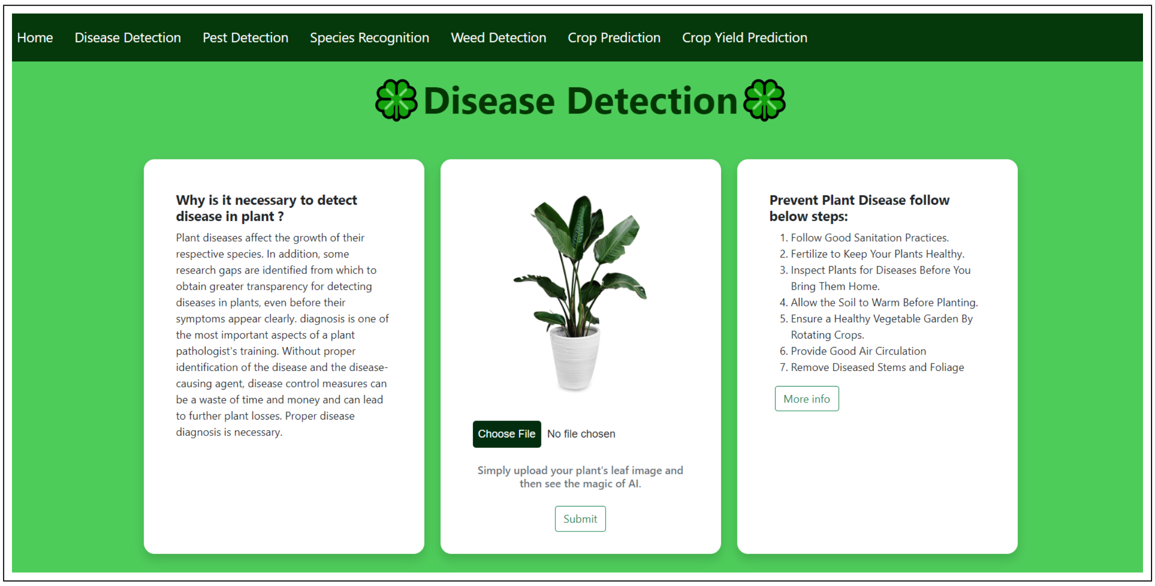


Figure 5.9: Disease Detection Homepage

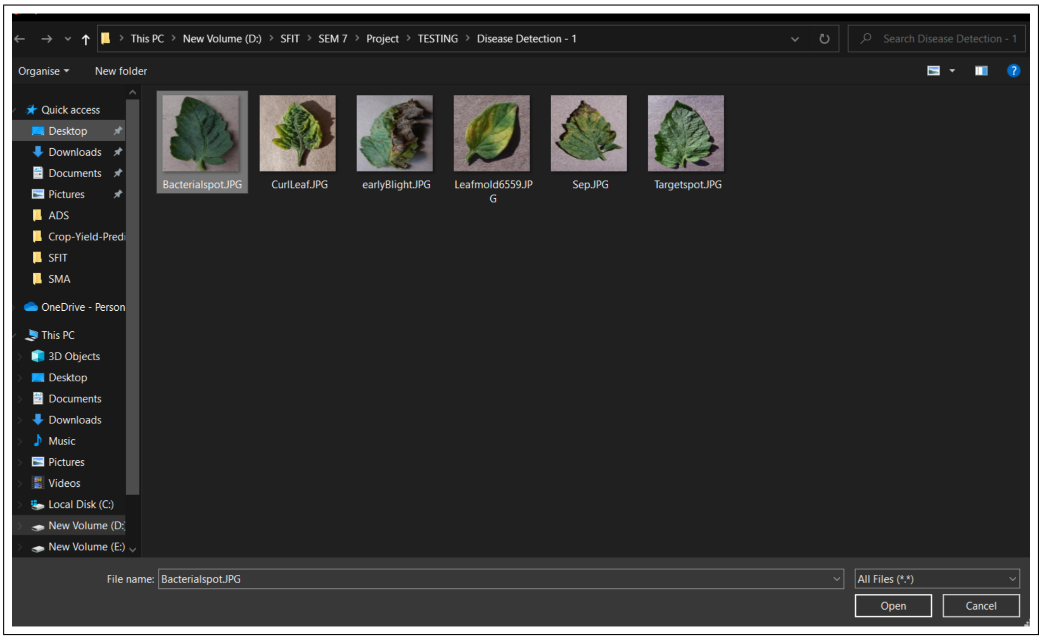


Figure 5.10: Selection of Infected Leaf Input

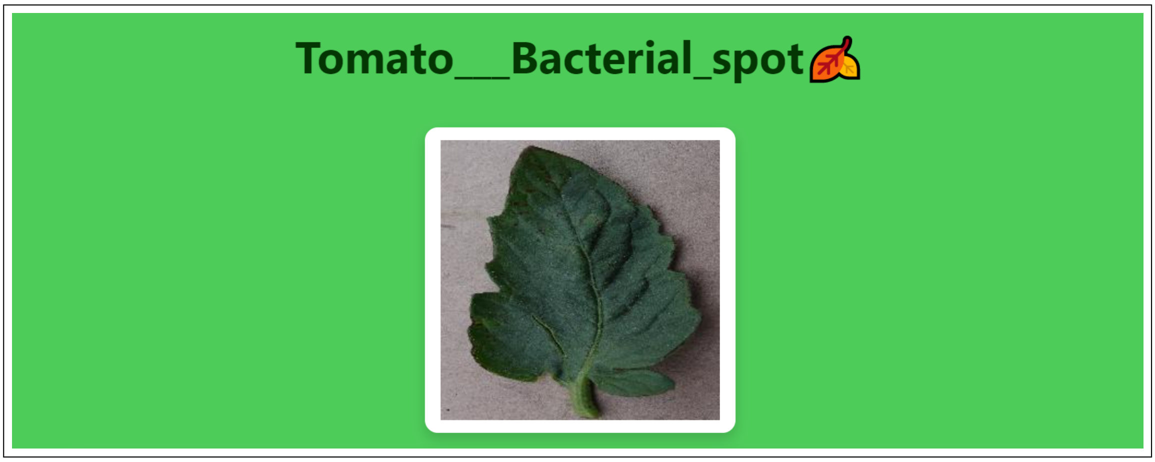


Figure 5.11: Disease Detection Output

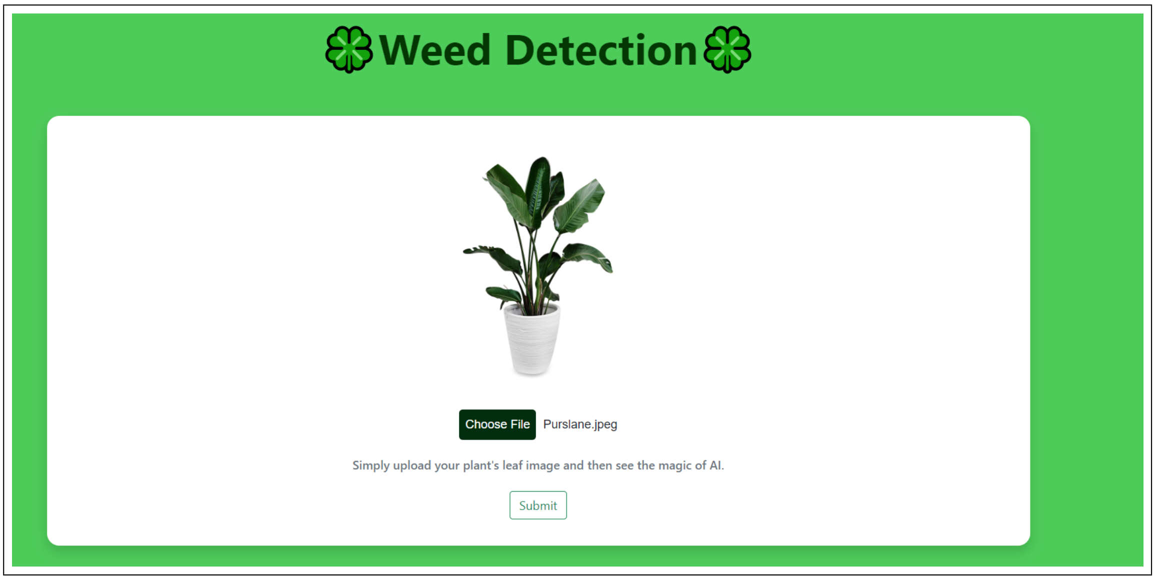


Figure 5.12: Weed Detection Homepage

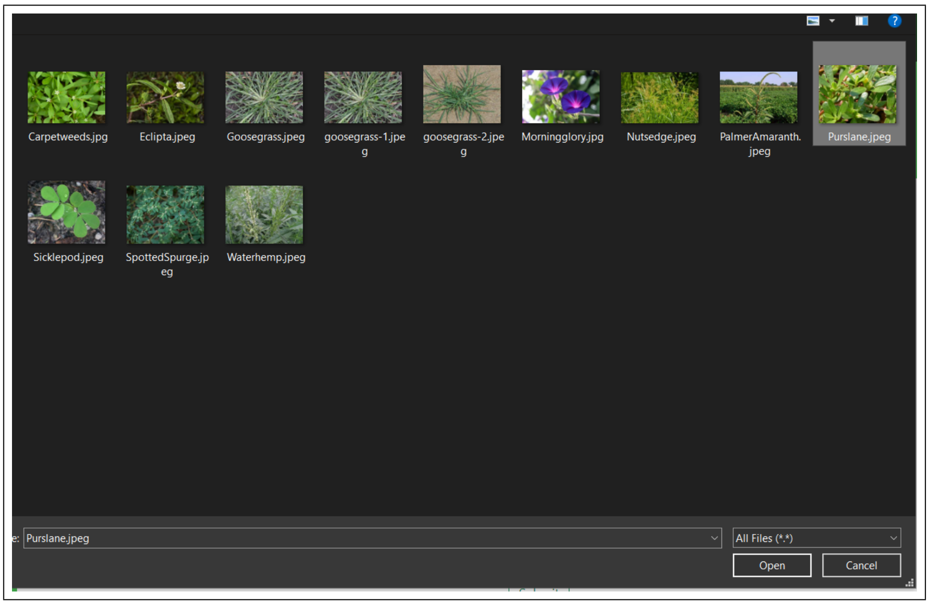


Figure 5.13: Selection of Weed Input

Figure 5.14: Weed Detection Output

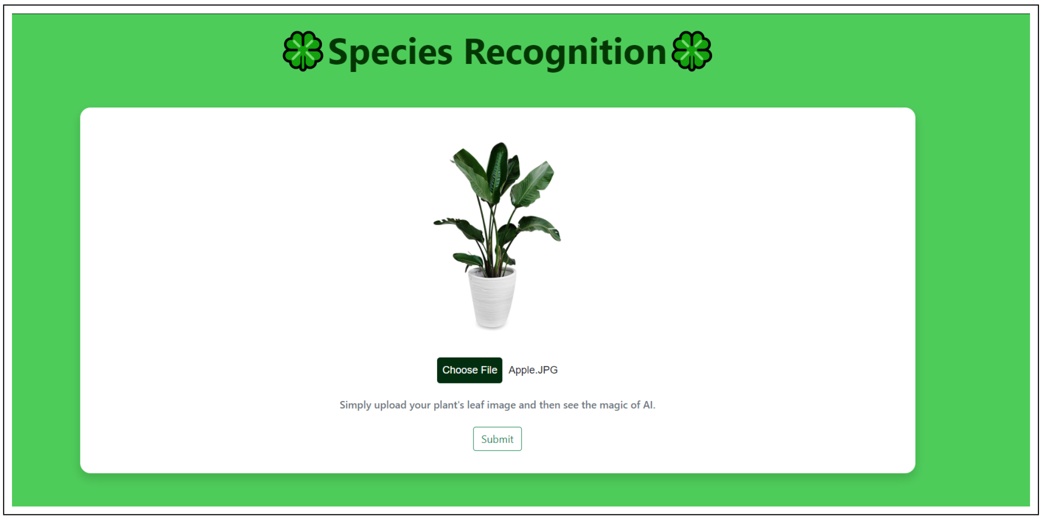


Figure 5.15: Species Recognition Homepage

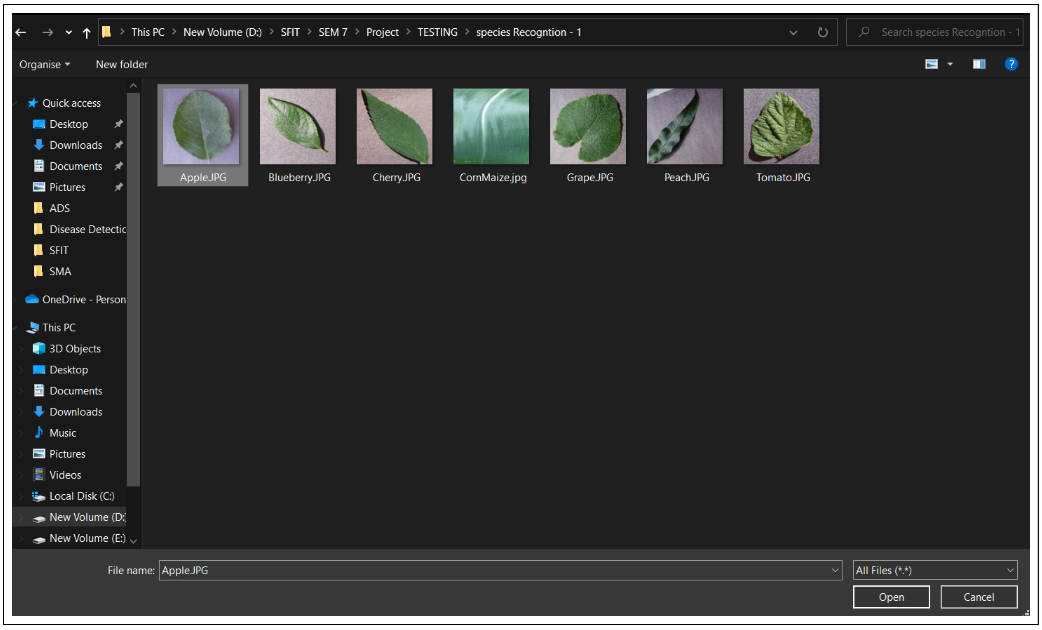


Figure 5.16: Selection of Leaf Input

Figure 5.17: Species Recognition Output

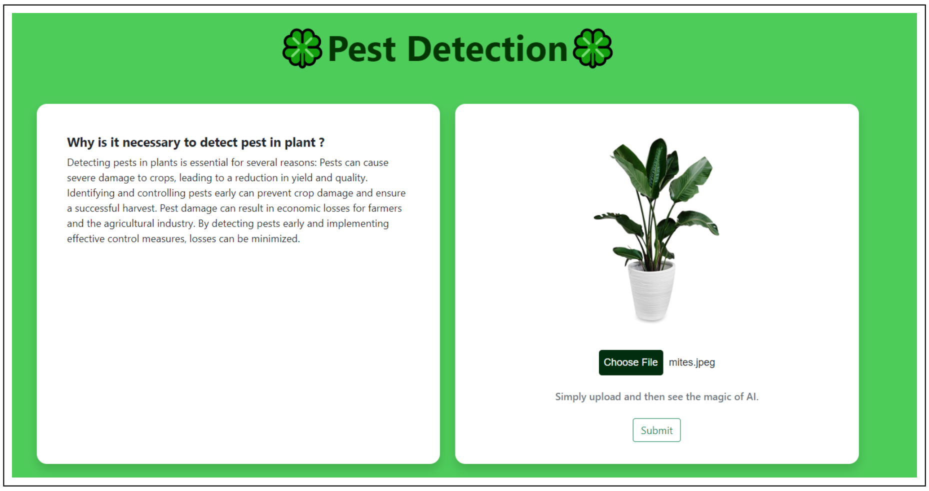


Figure 5.18: Pest Detection Homepage

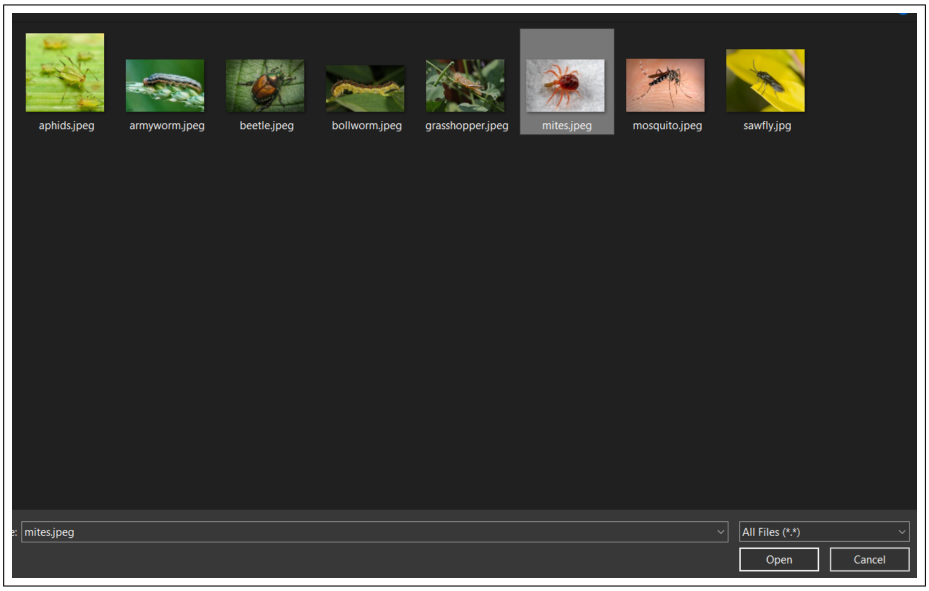


Figure 5.19: Selection of Pest Input

Figure 5.20: Pest Detection Output

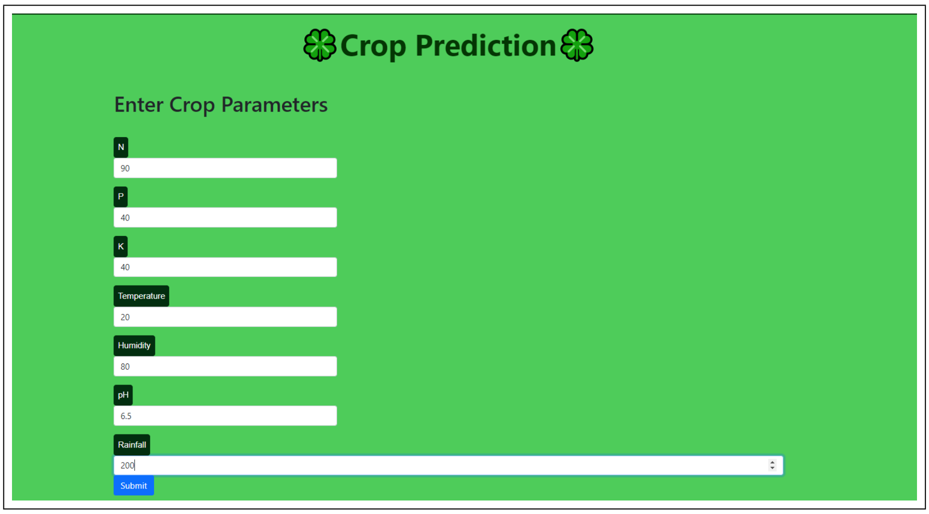


Figure 5.21: Crop Recommendation Input

Figure 5.22: Crop Recommendation Output

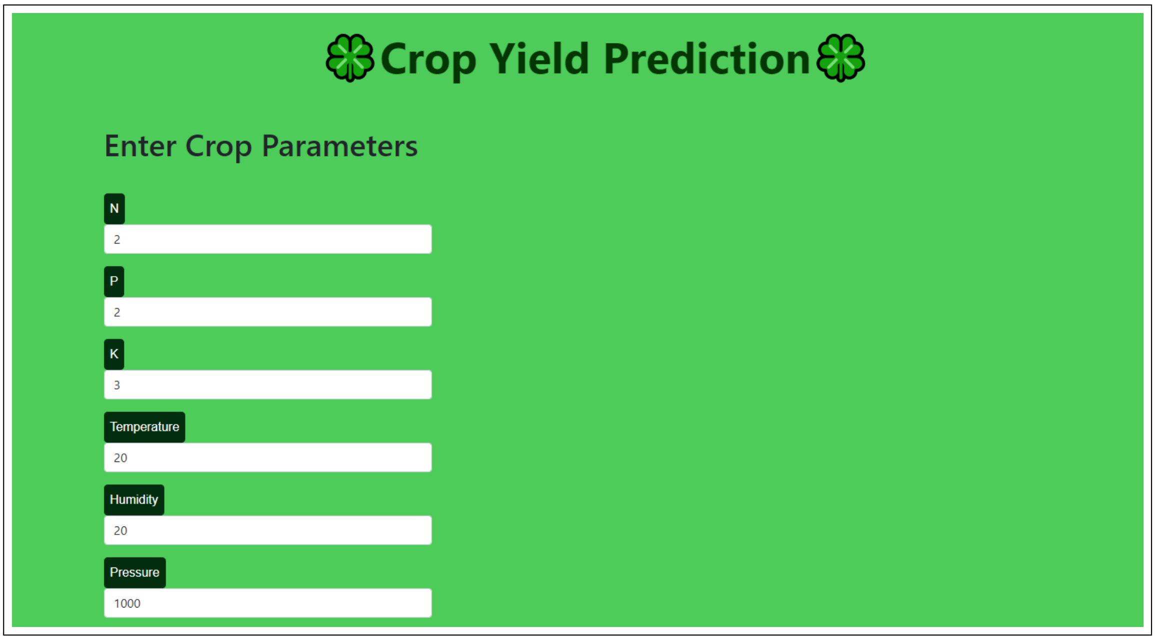


Figure 5.23: Crop Yield Precdiction Input

Figure 5.24: Crop Yield Prediction Input

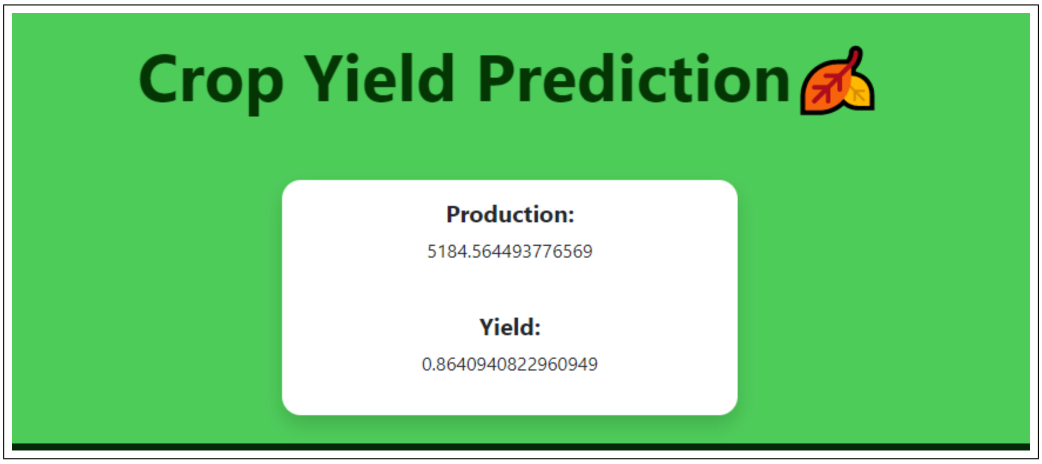


Figure 5.25: Crop Yield Prediction Output

Chapter 6

Implementation

6.1 Existing: Convolutional Neural Network for Classification of Crop Species and Disease, Weed, and Pest

6.1.1 Image Pre-processing

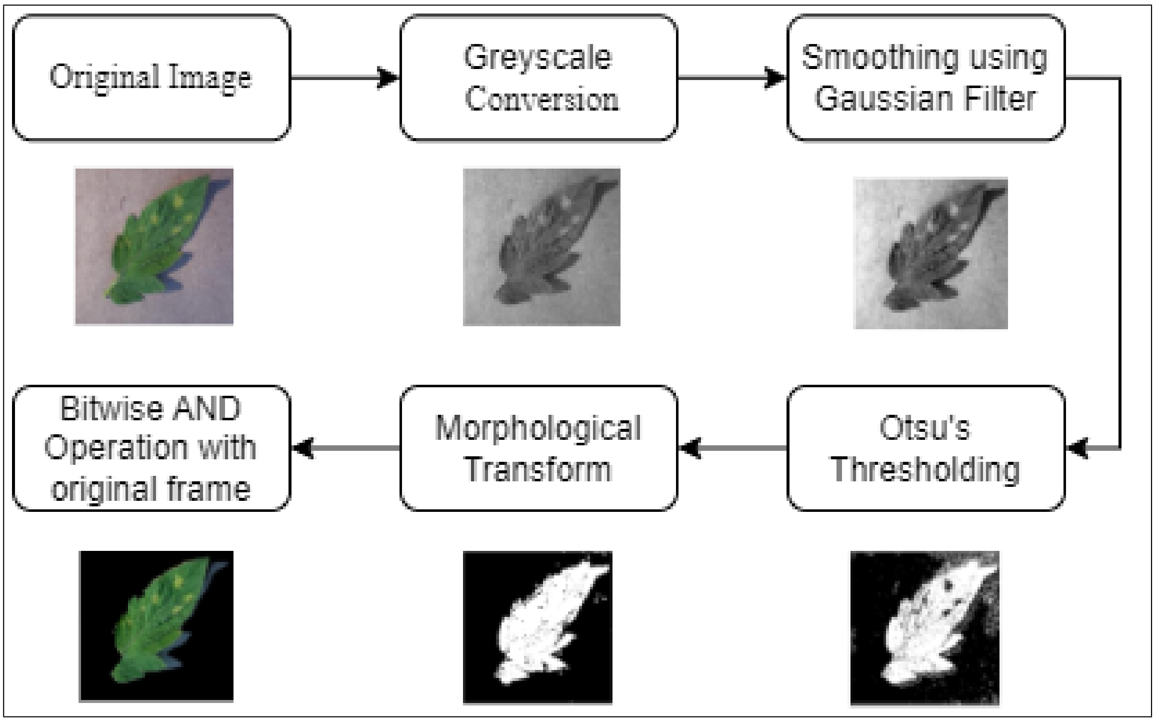


Figure 6.1: Stages of Image Preprocessing

Before using the dataset for detection and classification, the dataset images were preprocessed and segmented to obtain better accuracy. The image undergoes multiple stages of preprocessing as shown in Fig. 8. In the first stage, the background noise should be removed because of which the original RGB image is converted into a Greyscale image. In the second stage, the Gaussian filter is used to smoothen the image and thus eliminate all the noise in the image. The next stage is where we apply Otsu's Thresholding where the separation of foreground pixels is done from the background pixels. Then, the image undergoes morphological transformation to close the holes in the foreground. In the final stage, the Bitwise AND Operation is performed between the original image and the result of morphological transform. An image is obtained which has only the main image content without a background. This significantly improves the accuracy in detection and classification of images.

6.1.2 Working of CNN Resnet50 Code:

i) Architecture of ResNet50

```
from keras.applications import ResNet50


resnet = ResNet50(weights='imagenet',
include_top=False,
input_shape=(224, 224, 3))


for layer in resnet.layers[:-4]:
    layer.trainable = False


def nvidia_model():
    model = Sequential()
    model.add(resnet)


    #comment out
   # model.add(MaxPooling2D(pool_size=(2, 2)))
   # model.add(Dropout(0.5))


    model.add(Flatten())




    model.add(Dense(512, activation='relu'))
    model.add(Dropout(0.2))
```

```
    model.add(Dense(7,activation='softmax'))
    optimizer = Adam(lr=1e-3)
    model.compile(loss=
    'categorical_crossentropy', optimizer=optimizer,
    metrics=['accuracy'])

    return model

model = nvidia_model()
print(model.summary())
```

ii) Model Training

```
checkpoint = ModelCheckpoint('model.h5', monitor='val_loss',
verbose=1,
save_best_only=True,
mode='auto')

history = model.fit_generator(
    aug.flow(x_train, y_train, batch_size=BS),
    validation_data=(x_test, y_test),
    steps_per_epoch=len(x_train) // BS,
    epochs=10, verbose=1,
    callbacks=[checkpoint]
    )
```

iii) Prediction

```
from sklearn.metrics import classification_report
pred = model.predict(x_test, batch_size=32, verbose=1)
```

```
predicted = np.argmax(pred, axis=1)
report = classification_report(np.argmax(y_test, axis=1), predicted)
print(report)
def predict_weed(image_path):
    image_array = convert_image_to_array(image_path)
    np_image = np.array(image_array, dtype=np.float32)
    print(np_image.dtype)
    np_image = np.expand_dims(np_image,0)
    print(np_image.shape)
    plt.imshow(plt.imread(image_path))
    result = model.predict_classes(np_image)
    print(result)
    print((label_binarizer.classes_[result]))
```

6.1.3 Working of CNN DenseNet Code:

i) Architecture code for DenseNet:

```
def build_densenet():
    densenet = DenseNet121(weights='imagenet', include_top=False)

    input = Input(shape=(SIZE, SIZE, N_ch))
    x = Conv2D(3, (3, 3), padding='same')(input)

    x = densenet(x)

    x = GlobalAveragePooling2D()(x)
    x = BatchNormalization()(x)
    x = Dropout(0.5)(x)
    x = Dense(256, activation='relu')(x)
    x = BatchNormalization()(x)
    x = Dropout(0.5)(x)
```

```
    # multi output
    output = Dense(6,activation = 'softmax', name='root')(x)

    # model
    model = Model(input,output)

    optimizer = Adam(lr=0.002, beta_1=0.9,
    beta_2=0.999, epsilon=0.1,
    decay=0.0)
    model.compile(loss='categorical_crossentropy', optimizer=optimizer,
    metrics=['accuracy'])
    model.summary()
    return model
```

ii) Training phase for DenseNet:

```
model = build_densenet()
annealer = ReduceLROnPlateau(monitor='val_accuracy', factor=0.5,
patience=5, verbose=1,
min_lr=1e-3)
checkpoint = ModelCheckpoint('model.h5',
verbose=1, save_best_only=True)
# Generates batches of image data with data augmentation
datagen = ImageDataGenerator(rotation_range=360,
# Degree range for random rotations
                        width_shift_range=0.2,
                        # Range for random horizontal shifts
                        height_shift_range=0.2,
                        # Range for random vertical shifts
                        zoom_range=0.2,
                        # Range for random zoom
                        horizontal_flip=True,
                        # Randomly flip inputs horizontally
```

```
                         vertical_flip=True) #
                         Randomly flip inputs vertically

datagen.fit(x_train)
# Fits the model on batches with real-time data augmentation
hist = model.fit_generator(datagen.flow(x_train,
y_train,
batch_size=BATCH_SIZE),
               steps_per_epoch=x_train.shape[0]
               // BATCH_SIZE,
               epochs=EPOCHS,
               verbose=2,
               callbacks=[annealer, checkpoint],
               validation_data=(x_test, y_test))
```

6.2 Proposed: Bins Approach for Classification of Crop Species and Disease, Weed and Pest

6.2.1 Introduction

- We attempt to address the drawbacks of poor feature engineering and the large computational complexity of a traditional CNN (that have been discussed in the Results and Discussions chapter) by utilizing another technique that can extract essential content features pertaining to an improvised accuracy of the classification of the image.

6.2.2 Bins Approach

- In image processing, every color histogram depicts the total tonal distribution that occurs using a set of bins. With a traditional global color histogram approach, an image is usually encoded with its color histogram, and the distance between two images will be determined by the distance between their color histograms.

- However, this technique uses 256 bins as a feature vector and increases the size, time, and space complexity required for classification. Although the presence of most of these features can be ignored, it is still a crucial factor to be considered while designing an efficient CBIR system.

- In our system we have resolved this issue by exploring a novel technique to form these bins so that color details of the image will be separated properly, and feature vector size can be reduced through dimensionality reduction without disturbing the features.

- Using the Bins Approach, we are focusing on the use of image histograms to extract the image features and also trying to reduce the size of the feature vector.

6.2.3 Algorithm View with Implementation Detail

With respect to the algorithms of the Bins Approach mentioned in Chapter 5, we shall now demonstrate each and every process of the Bins Approach using the following sample Leaf Images:

Figure 6.2: Sample images of Tomato Leaf Mold and Tomato Target Spot

6.2.4 Demonstration

i) Image Preprocessing

Image preprocessing is the application of a set of techniques and algorithms to a digital image to analyze, enhance, or optimize image characteristics such as sharpness and contrast. In this step, we are removing the noise from the cell image.

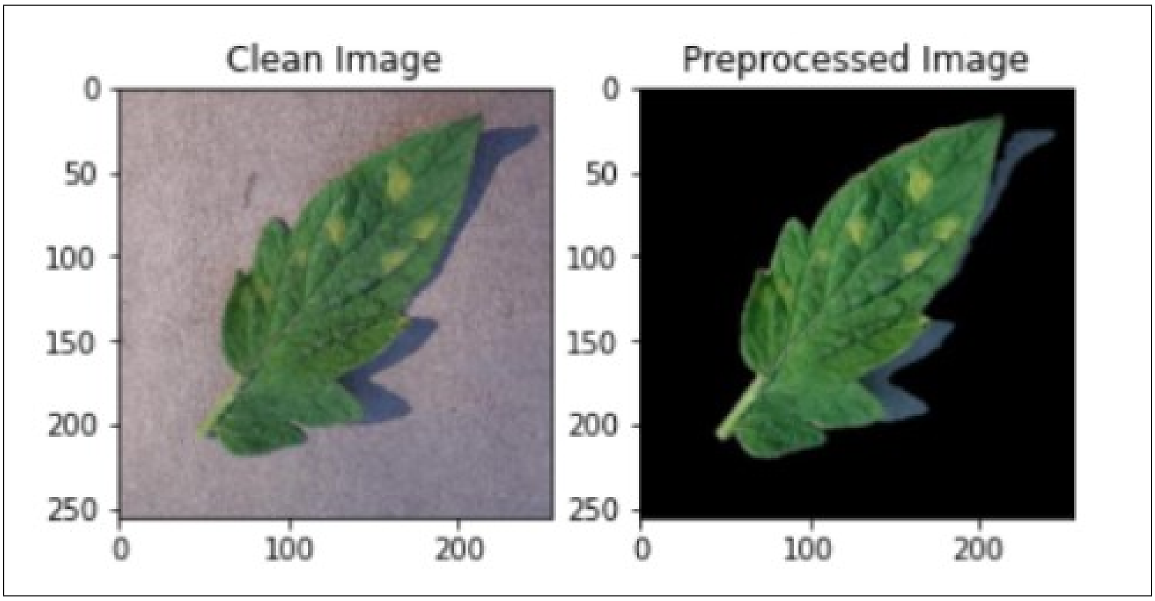


Figure 6.3: Sample input image and preprocessed image

ii) Deducing Bins Value

With Respect to the algorithm:

Step 1: Spilt the image into R, G and B planes.

Step 2: Obtain the histogram for each plane.

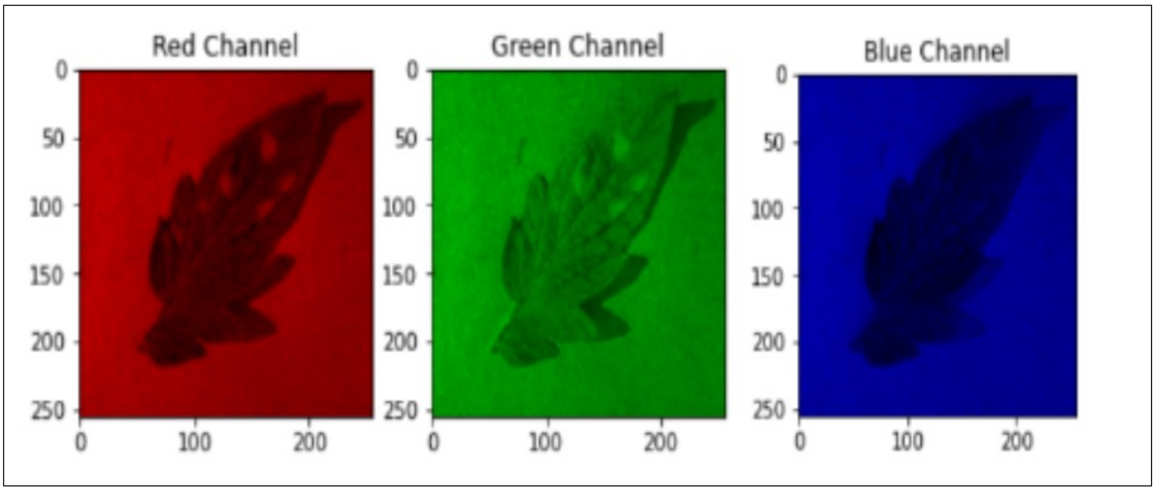


Figure 6.4: Sample leaf image separated into R, G and B planes

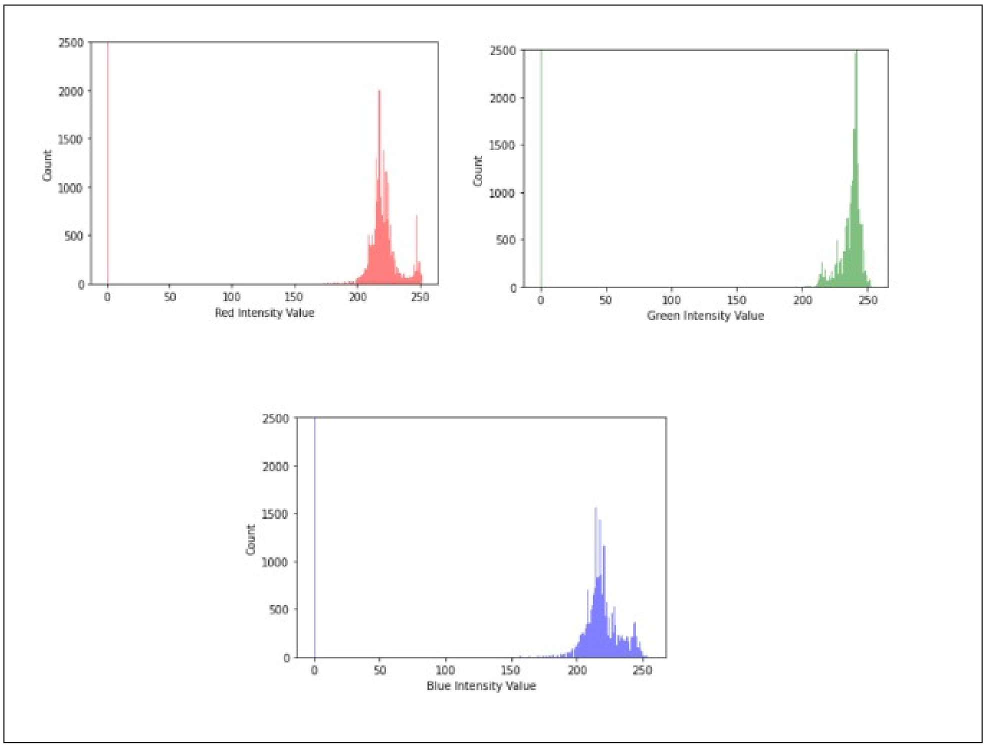


Figure 6.5: Histograms for R, G and B planes

Table 6.1: Synopsis of total CG values for sample leaf image

CG of R-plane	CG of G-plane	CG of B-plane
69.6586	74.6746	69.1615

Step 3: Divide each histogram into 2 parts and assign a unique flag to each part (either '0' OR '1').

- We have focused on the Center of Gravity based partitioning to divide each histogram into two parts. Fig 6.6 shows the partitioning of the histogram in two parts Linear Partitioning and CG.

- When we apply LP, we notice that every color histogram does not give balanced partitioning; instead, it shows a disfigured one. This is basically because, it considers only the count of pixels into consideration, and ignores the intensity values.

- However, with the computation of the CG threshold in our project, we are giving equal importance to the pixels as well as their intensities.

- Each of these intensities is considered as weight of the pixels so that we divide these pixels into two values according to their weights by computing the CG.

- This idea is exactly what led us to label these partitions and identify them as '0' and '1', such that both of these partitions will have the same intensity. This process helps in showing an improvement in the retrieval performance of the system

- The formula to compute a CG is given by:

$$\text{CG} = \frac{(L_i W_i + L_{i+1} W_{i+1} + ... + L_n W_n)}{\sum_{i=1}^{n} W_i} \qquad (1)$$

Step 4: To extract the color feature of the image, pick up the original image pixel and check its R, G and B values find out in the histogram that in which range these values exactly falls, based on it assign the unique flags to the r, g and b values of that pixel with respect to the partition of the histogram it belongs.

- In this case, the unique flag is the CG that we computed in the previous step for every channel.

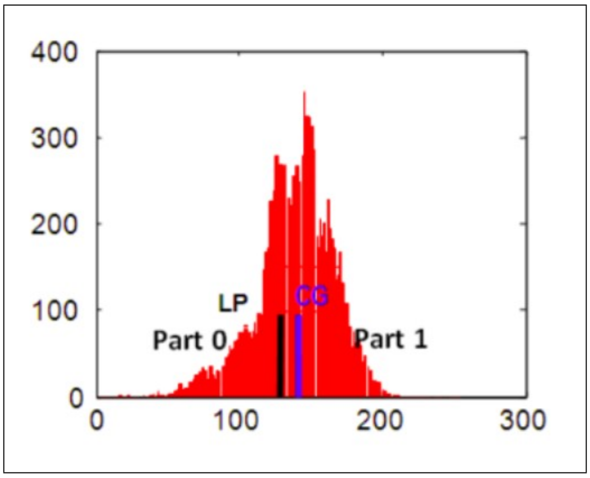


Figure 6.6: Sample Partitioning using Centre of gravity[20]

- The height and the width of the image have already been pre-computed through Python functions.
- To understand this concept better, imagine a virtual sliding box that starts from the topmost left pixel of the image, and slide it across every pixel along the horizontal line.
- If it reaches the rightmost pixel of that line, start performing the sliding operation from the leftmost pixel just below the one where it started from.
- Whilst performing this operation, calculate the intensity of the pixel corresponding to the respective R, G, and B plane, as well as keep the CG of the respective channels ready.
- At every looping iteration, check if the intensity of the pixel of the corresponding channel is lesser than or equal to the respective CG pertaining to that channel. If yes, add a '0', else '1'.

Step 5: Count of pixels in the bin: Based on the flags assigned to each pixel with respect to the R, G B values and 2 partitions (e. g. 0 and 1) of the histogram we can have 8 combinations from 000 to 111 which are the total 8 bins following and implementing above algorithm we get 8 bins with count of Pixels in each of the 8 bins with respect to the image under feature extraction process.

- We keep the computed CG matrix ready for every channel with respect to the previous step. We also keep 8 bins of corresponding binary values ready (ie:

from 000=0 to 111=7).

- Again, like we did in the previous step, imagine a virtual sliding box that starts from the topmost left pixel of the image, and slide it across every pixel along the horizontal line. However, unlike the previous step, we perform this sliding operation along three computed CG matrices in a single iteration/loop.

- Since the CG matrices are the same as the height and the width of the original input image, no pixels are lost, and this can be proved from Table 10 described below for each sample leaf images.

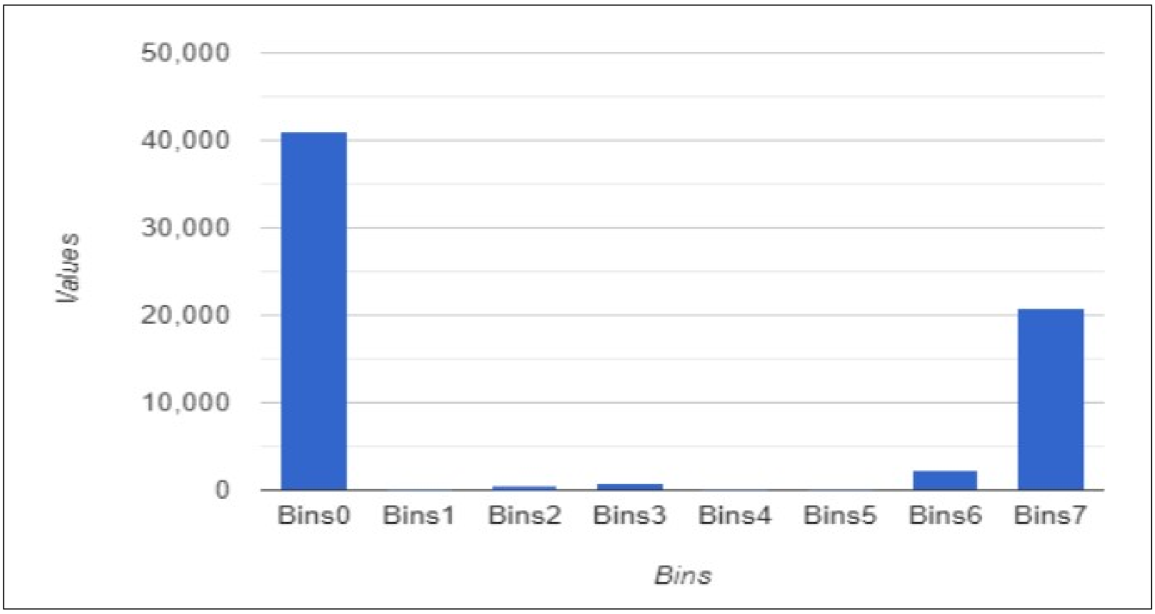


Figure 6.7: Feature Vector representation in the form of 8 bins.

- Upon cross-verification of these bin values by adding their total count as well as multiplying the height and the width of the original respective sample images, we are given to accept that the code is fundamentally and logically accurate.

iii) Calculating Bin Values for Multiple Images:

- Since we already calculated the bin feature of the leaf images for the sample classes, we pass a loop function wherein we acquire 900 clean image samples each for the plant disease classes.

- These images are then passed through the entire bins process to produce the values that are written on a .csv file to produce a dataset sample cell images.

Bins0	Bins1	Bins2	Bins3	Bins4	Bins5	Bins6	Bins7
23079	626	5978	1891	2021	2692	840	33779
623999	12106	84203	2381	22349	42763	54300	779257
1243145	164224	560135	17079	191616	362074	80069	1244856
2206780	50819	605944	440099	457919	367326	52941	2577752
23891	262	3010	503	769	410	312	27993
6189	63	1265	336	157	150	668	8137
20558	181	2446	294	598	520	220	24791
35140	695	5588	361	1047	2307	5706	49960
11372	185	3068	149	409	839	870	18198
15152	201	1848	115	296	683	2343	21548

Figure 6.8: Bin feature vector for Multiple Images.

6.2.5 Implementation of Bins Approach

1. Import all the Python library functions and display the image:

```
from skimage.color import rgb2gray
import numpy as np
import cv2
import matplotlib.pyplot as plt
%matplotlib inline
from scipy import ndimage
import skimage.io as io
from copy import deepcopy
image = plt.imread(r'C:\Users\USER\Desktop\b.png')
image.shape
plt.imshow(image)
```

2. Remove the noise from the image using appropriate preprocessing techniques::

```
import numpy as np
import cv2
from matplotlib import pyplot as plt
img = cv2.imread(r'C:\Users\USER\Desktop\b.png')
```

```
dst = cv2.fastNlMeansDenoisingColored(img,None,10,10,7,29)
plt.subplot(121),plt.imshow(img),plt.title('Noisy image')
plt.subplot(122),plt.imshow(dst),plt.title('Clean image')
plt.show()
cv2.imwrite(r'C:\Users\USER\Desktop\clean.png', dst)
```

3. Split the image into three channels, i.e. Red, Green and Blue Channels:

```
img=io.imread(r'C:\Users\USER\Desktop\clean.png')
r=deepcopy(img)
g=deepcopy(img)
b=deepcopy(img)
r[:,:,1]=0
r[:,:,2]=0
g[:,:,0]=0
g[:,:,2]=0
b[:,:,0]=0
b[:,:,1]=0
fig, ax=plt.subplots(ncols=2,nrows=2)
ax[0,0].imshow(img)
ax[0,0].set_title('original')
ax[0,1].imshow(r)
ax[0,1].set_title('red channel')
ax[1,0].imshow(g)
ax[1,0].set_title('green channel')
ax[1,1].imshow(b)
ax[1,1].set_title('blue channel')
plt.show()
```

4. Compute histogram for Red Channel (Similarly for Green, Blue Channel):

```
from skimage import io
```

```
import matplotlib.pyplot as plt
image = io.imread(r'C:\Users\USER\Desktop\clean.png')
_ = plt.hist(image[:, :, 0].ravel(), bins = 256, color =
'red', alpha = 0.5)
_ = plt.xlabel('Red Intensity Value')
_ = plt.ylabel('Count')
plt.show()
x = np.array(image[:,:,0], dtype = np.uint8)
cv2.imwrite(r'C:\Users\USER\Desktop\rplane.png', x)
plt.imshow(x)
```

5. Compute the pixel count for every color histogram of the respective channel:

```
image = io.imread(r'C:\Users\USER\Desktop\rplane.png')
a = np.array(image[:,:,], dtype = np.uint8)
print("MATRIX FOR R PLANE HISTOGRAM WITH COUNTS:")
#a.flatten()
#print (a)
#for i in a.flatten():
# print(i)
d = dict()
for i in range (0,256):
 d[i]=0
x = []
for i in range (0,256):
 x.append(i)
#print(x)
y = []
for i in range (0,256):
 y.append(d[i])
#print(y)
num = 0
```

```
den = 0
for i in a.flatten():
 d[i] = d.get(i,0) + 1
for i in x:
 b = d.get(i,0) + 1

 num = num + b*i
 den = den + b
cogr = num / den
print (d)
print("COG of R plane=",cogr)
```

6. Partition every channel using CG and compute the matrix:

```
print("%%%%%%%%%%%%%%% Partition of the R matrix using COG
%%%%%%%%%%%%%%%")
image1 = io.imread(r'C:\Users\USER\Desktop\rplane.png')
h, w = image1.shape
for r in range(h):
 for c in range(w):
 if image1[r, c] <= cogr:
 image1[r, c] = 0
 else:
 image1[r, c] = 1
print(image1)
```

9. With the matrix computed, compute the r, g and the b value of the pixel at that point and add it to the assigned bin corresponding to the binary value

```
bins0 = 0
bins1 = 0
bins2 = 0
bins3 = 0
```

```
bins4 = 0
bins5 = 0
bins6 = 0
bins7 = 0
for i in range(h):
 for j in range(w):
 r= image1[i,j]
 g= image2[i,j]
 b= image3[i,j]

 if r==0 and g==0 and b==0:
 bins0= bins0 + 1
 elif r==0 and g==0 and b==1:
 bins1= bins1 + 1
 elif r==0 and g==1 and b==0:
 bins2= bins2 + 1
 elif r==0 and g==1 and b==1:
 bins3= bins3 + 1
 elif r==1 and g==0 and b==0:
 bins4= bins4 + 1
 elif r==1 and g==0 and b==1:
 bins5= bins5 + 1
 elif r==1 and g==1 and b==0:
 bins6= bins6 + 1
 elif r==1 and g==1 and b==1:
 bins7= bins7 + 1
print("Bins 0:",bins0)
print("Bins 1:",bins1)
print("Bins 2:",bins2)
print("Bins 3:",bins3)
print("Bins 4:",bins4)
print("Bins 5:",bins5)
```

```
print("Bins 6:",bins6)
print("Bins 7:",bins7)
```

10. Splitting data into train and test data

```
dataset = pd.read_csv('disease_detection.csv')
X = dataset.iloc[:, :-1].values
y = dataset.iloc[:, -1].value

from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test =
train_test_split(X, y,
test_size = 0.25,
random_state = 0)
```

11. Classification using SVM

```
from sklearn.svm import SVR

from sklearn.preprocessing import StandardScaler
sc_X = StandardScaler()
X_train = sc_X.fit_transform(X_train)
X_test = sc_X.transform(X_test)
from sklearn import svm
svm1 = svm.SVC(kernel='linear', C = 0.01)
svm1.fit(X_test,y_test)
```

12.Classification using Decision Tree

```
from sklearn.tree import DecisionTreeClassifier
classifier = DecisionTreeClassifier(criterion =
'entropy',
random_state = 0)
classifier.fit(X_train, y_train)
```

13. Classification using Logistic Regression

```
from sklearn.linear_model import LogisticRegression
classifier = LogisticRegression(random_state = 0)
classifier.fit(xtrain, y_train
```

6.3 Crop Yield Prediction

6.3.1 Introduction

In this module, we explore the prediction of crop yield using various machine learning techniques. The dataset used in this study consists of several features such as area, temperature, wind speed, pressure, humidity, soil type, N, P, K, state names, district names, season names, and crop names. The primary objective of this module is to predict the yield of crops using Random Forest Regressor, Polynomial Support Vector Machine, and XGBRegressor. The yield is calculated as the ratio of production and area.

6.3.2 Implementation of Crop Yield Prediction

i) Reading Data

```
import pandas as pd
df = pd.read_csv('Crop Yield Prediction/finalised_dataset.csv',na_values='=')
df.info()
df.columns
```

ii) Reducing Data to One State for Ease

```
df = df[df['state_names'] == "Maharashtra"]
df.info()
```

iii) Corelation Heatmap

```
import matplotlib.pyplot as plt
import seaborn as sb
```

```
C_mat = df.corr()
fig = plt.figure(figsize = (15,15))

sb.heatmap(C_mat, vmax = .8, square = True)
plt.show()
```

iv) Restricting Data Analysis to Post-2004 Records for Improved Accuracy and Reliability

```
df = df[df['crop_year']>=2004]
df
```

v) Converting Categorical Data to Numerical Form

```
df = df.join(pd.get_dummies(df['district_names']))
df = df.join(pd.get_dummies(df['season_names']))
df = df.join(pd.get_dummies(df['crop_names']))
df = df.join(pd.get_dummies(df['state_names']))
df = df.join(pd.get_dummies(df['soil_type']))
df
```

vi) Calculating Crop Yield and Creating a New Column

```
df['Yield'] = df['production']/df['area']
df
```

vi) Removing Unnecessary Columns

```
df=df.drop('district_names', axis=1)
df = df.drop('season_names',axis=1)
df = df.drop('crop_names',axis=1)
df = df.drop('crop_year', axis=1)
df = df.drop('state_names', axis=1)
df = df.drop('soil_type', axis=1)
```

vi) Scaling the 'area' Feature using Min-Max Normalization

```
x = df[['area']].values.astype(float)


min_value = x.min()
print(min_value)
max_value = x.max()
print(max_value)
x_scaled = (x - min_value) / (max_value - min_value)


df['area'] = x_scaled
df
```

vii) Filling Missing Values with Mean

```
df = df.fillna(df.mean())
df
```

viii) Splitting Data into Training and Testing Sets using Scikit-Learn

```
from sklearn.model_selection import train_test_split
a = df
b = df['Yield']
features_list=['area', 'temperature', 'wind_speed', 'pressure',
       'humidity', 'N', 'P', 'K', 'AHMEDNAGAR', 'AKOLA', 'AMRAVATI',
       'AURANGABAD', 'BEED', 'BHANDARA', 'BULDHANA', 'CHANDRAPUR', 'DHULE',
       'GADCHIROLI', 'GONDIA', 'HINGOLI', 'JALGAON', 'JALNA', 'KOLHAPUR',
       'LATUR', 'NAGPUR', 'NANDED', 'NANDURBAR', 'NASHIK', 'OSMANABAD',
       'PALGHAR', 'PARBHANI', 'PUNE', 'RAIGAD', 'RATNAGIRI', 'SANGLI',
       'SATARA', 'SINDHUDURG', 'SOLAPUR', 'THANE', 'WARDHA', 'WASHIM',
       'YAVATMAL', 'Kharif ', 'Rabi', 'Summer ', 'Whole Year',
       'Arhar/Tur', 'Bajra', 'Castor seed', 'Cotton(lint)', 'Gram',
       'Groundnut', 'Jowar', 'Linseed', 'Maize', 'Moong(Green Gram)',
       'Niger seed', 'Other  Rabi pulses', 'Other Cereals & Millets',
```

```
        'Other Kharif pulses', 'Ragi',
        'Rapeseed &Mustard', 'Rice', 'Safflower',
        'Sesamum', 'Soyabean', 'Sugarcane', 'Sunflower', 'Tobacco', 'Urad',
        'Wheat', 'other oilseeds', 'Maharashtra', 'chalky', 'clay', 'loamy',
        'peaty', 'sandy', 'silt', 'silty']

a=df[features_list]
a_train, a_test, b_train, b_test = train_test_split(a, b,
test_size = 0.3, random_state = 42)
```

ix) Implementing Random Forest Regressor and Evaluating Performance using Scikit-Learn

```
from sklearn.ensemble import RandomForestRegressor
regr = RandomForestRegressor(max_depth=2, random_state=0, n_estimators=100)
regr.fit(a_train, b_train)
b_pred = regr.predict(a_test)

from sklearn.metrics import mean_squared_error as mse
from sklearn.metrics import mean_absolute_error as mae
from sklearn.metrics import r2_score

print('MSE =', mse(b_pred, b_test))
print('MAE =', mae(b_pred, b_test))
print('R2 Score =', r2_score(b_pred, b_test))
```

x) Implementing Polynomial Support Vector Machine Regressor and Evaluating Performance using Scikit-Learn

```
from sklearn.svm import SVR
regressorpoly=SVR(kernel='poly',epsilon=1.0)
regressorpoly.fit(a_train,b_train)
pred=regressorpoly.predict(a_test)
print(regressorpoly.score(a_test,b_test))
```

```
print("R2:",r2_score(b_test,pred))

from sklearn.metrics import mean_absolute_error, mean_squared_error

# calculate MAE and MSE
mae = mean_absolute_error(b_test, pred)
mse = mean_squared_error(b_test, pred)

# print the results
print("MAE:", mae)
print("MSE:", mse)
```

xi) Implementing XGBoost Regressor and Evaluating Performance using Scikit-Learn

```
from xgboost import XGBRegressor
from sklearn.metrics import mean_absolute_error
XGBModel = XGBRegressor()
XGBModel.fit(a_train,b_train , verbose=False)

# Get the mean absolute error on the validation data :
XGBpredictions = XGBModel.predict(a_test)
MAE = mean_absolute_error(b_test , XGBpredictions)
print('XGBoost validation MAE = ',MAE)
XGBpredictions
```

6.4 Crop Recommendation

6.4.1 Introduction

This module uses a set of pre-selected features such as 'N', 'P', 'K', 'temperature', 'humidity', 'ph', 'rainfall', and a list of 22 crops such as rice, maize, chickpea, and more to recommend the best crop to grow in a particular area. In this module, three different classification models: Decision Tree,Naive Bayes, Random Forest, and SVM will be used to make crop recommendations based on the given input features.

6.4.2 Implementation of Crop Recommendation

i) Loading and Viewing Crop Recommendation Dataset

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
crop_data=pd.read_csv("Crop_recommendation.csv")
crop_data
```

i) Loading and Viewing Crop Recommendation Dataset

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
crop_data=pd.read_csv("Crop_recommendation.csv")
crop_data
```


ii) Checking the Data Shape and Information

```
crop_data.shape
crop_data.info()
```

iii) Changing the Name of label to Crop for Readability

```
crop_data.rename(columns = {'label':'Crop'}, inplace = True)
crop_data
```

iv) Statistical Inference of the Dataset

```
crop_data.describe()
```

v) Checking Missing Values

```
crop_data.isnull().sum()
```

vi) Visualizing the features

```
ax = sns.pairplot(crop_data)
ax
```

vii) Visualizing Correlation Matrix Using Heatmap

```
crop_data.corr()
sns.heatmap(crop_data.corr(), annot =True)
plt.title('Correlation Matrix')
```

viii) Shuffling the Dataset to Remove Order

```
from sklearn.utils import shuffle
df  = shuffle(crop_data,random_state=5)
df.head()
```

ix) Selection of Feature and Target variables

```
x = df[['N', 'P','K','temperature', 'humidity', 'ph', 'rainfall']]
target = df['Crop']
```

x) Encoding Target Variable

```
y = pd.get_dummies(target)
y
```

xi) Selection of Feature and Target variables

```
x = df[['N', 'P','K','temperature', 'humidity', 'ph', 'rainfall']]
target = df['Crop']
```

xii) Splitting Dataset

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.25,
random_state= 0)

print("x_train :",x_train.shape)
print("x_test :",x_test.shape)
print("y_train :",y_train.shape)
print("y_test :",y_test.shape)
```

xiii) Importing Necessary Libraries for Multi-Output Classification

```
# Importing necessary libraries for multi-output classification

from sklearn.datasets import make_classification
from sklearn.multioutput import MultiOutputClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.naive_bayes import GaussianNB
```

xiii) Training Gaussian Naive Bayes Classifier using MultiOutputClassifier

```
gnb = GaussianNB()
model = MultiOutputClassifier(gnb, n_jobs=-1)
model.fit(x_train, y_train)
```

xiv) Training Multi-Output Decision Tree Classifier

```
from sklearn.tree import DecisionTreeClassifier
```

```
clf = DecisionTreeClassifier(random_state=6)
multi_target_decision = MultiOutputClassifier(clf, n_jobs=-1)
multi_target_decision.fit(x_train, y_train)
```

xv) Training Multi-output Classification using Random Forest Classifier

```
forest = RandomForestClassifier(random_state=1)
multi_target_forest = MultiOutputClassifier(forest, n_jobs=-1)
multi_target_forest.fit(x_train, y_train)
```

Chapter 7

Testing

7.1 Test Cases

7.1.1 Testing Consists of the following types:

1. **Unit Testing:** In our project, unit testing was performed along with specific modules such as pre-processing, segmentation, feature extraction, and classification.

2. **Integration Testing:** In our project, integration testing was performed in between the modules mentioned in unit testing.

3. **System Testing:** In our project, system testing was performed to the entire classification system by clubbing and validating everything that was prepared to date.

4. **Acceptance Testing:** We have Performed Testing on Valid and Invalid input and also on unsupported operations.

7.1.2 Types of Testing used

Through the course of the project, we focused more on the implementation than the GUI. However, we planned on keeping the front end as simple as possible, as the maxim that we followed, was mainly to focus on Disease Detection, Pest Detection, Weed Detection, Species Recognition , Crop Recommendation and Crop Yield Prediction. We looked through various options, and for our interface, we finally chose to use Flask, a standard Web framework for Python. Flask provides a fast and easy way to create Web Applications applications. We chose simple widgets that were sufficient enough for our project, and performed the following testing cases summarized in Table 11.

Table 7.1: Test cases for Precision Agriculture

Test Case ID	Input Specification	Output Specification	Expected Output	Observed Output	Test Pass/Fail
1	Checking if images are inputted correctly	Input image is displayed	Display the crop image	Image displayed correctly	Pass
2	Checking if input crop disease image is classified correctly	Correct classification results displayed	Display proper details of the crop	Disease classified correctly	Pass
3	Checking if input crop species image is classified correctly	Correct classification results displayed	Display proper details of the crop	Species classified correctly	Pass
3	Checking if input weed species image is classified correctly	Correct classification results displayed	Display proper details of the weed	species classified correctly	Pass
4	Checking if the numeric parameters give correct recommendations	Correct crop recommendations	Display recommended crops	Correct crops recommended	Pass
5	Checking if numeric parameters give correct yield predictions	Correct yield predictions	Display yield predictions	Correct yield prediction	Pass
6	Checking if input pest image is classified correctly	Correct classification results displayed	Display proper details of the pest	Pest classified correctly	Pass

7.2 Performance Evaluation Parameters

The performance evaluation parameters that we are planning to use are:

- Accuracy:

  It is the proportion of the system's correct predictions for the overall number of predictions. It determines how often the classifier will show the correct output. The ideal value is 0.

$$\text{accuracy} = \frac{TP + TN}{TP + FN + TN + FP} \tag{2}$$

- F1 score:

  It is an indicator of the accuracy of a model on a dataset. It is used to test binary classification systems that classify examples as positive or negative. It is the harmonic mean of the precision and recall of the model. The ideal value is 1 ,and the worst case value is 0.

$$F1 = \frac{2 \times \text{ precision } \times \text{ recall}}{\text{precision } + \text{ recall}} \tag{3}$$

- Recall:

  As shown in equation no.3 Recall is calculated as the number of true positives divided by the total number of true positives and false negatives. The ideal value is 1,and the worst case value is 0.

$$\text{recall} = \frac{TP}{TP + FN} \tag{4}$$

- Precision:

  It is the proportion of True Positives divided by the number of True Positive and False Positive. The ideal value is 1 ,and the worst case value is 0.

$$\text{precision} = \frac{TP}{TP + FP} \tag{5}$$

- MAE (Mean Absolute Error):

  MAE measures the average absolute difference between the predicted and actual values of a set of observations. It is calculated as follows:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{6}$$

- MSE (Mean Squared Error):

  MSE measures the average squared difference between the predicted and actual values of a set of observations. It is calculated as follows:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \tag{7}$$

- R-squared (R2):

  R-squared is a statistical measure that represents the proportion of the variance in the dependent variable that is explained by the independent variable(s) in a linear regression model. It is also known as the coefficient of determination:

$$\text{R2} = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{8}$$

Chapter 8

Result and Discussions

8.1 Database Description:

8.1.1 Disease Detection and classification :

The Plant Village dataset[6] was created by a team of researchers at Penn State University, led by Dr. David Hughes.The dataset contains images of 14 crop species and 26 diseases, as well as healthy leaf images, captured under controlled conditions in a greenhouse. The images were collected using a variety of cameras and lighting conditions, to account for different real-world scenarios.

Table 8.1: Dataset for Disease detection and classification

	Tomato Bacterial spot	Tomato Early blight	Tomato Leaf Mold	Tomato Septoria leaf spot	Tomato Target Spot	Tomato Yellow Leaf Curl Virus
Training (80%)	800	800	800	800	800	800
Testing (20%)	200	200	200	200	200	200

8.1.2 Pest Prediction, Detection and Identification :

The Pest Dataset available on Kaggle is a collection of images that was created by SimranVolunesia. It consists of a total of 3,150 images, with 350 images for each of the 9 classes. The images are in .jpg format and the classes include aphids, armyworms, beetles, bollworms, grasshoppers, mites, mosquitoes, sawflies, and stem borers.

Table 8.2: Dataset for Pest Prediction, Detection & Identification (Part 1)

	Aphids	Army -worm	Beetle	Bollworm
Training (80%)	256	256	256	256
Testing (20%)	64	64	64	64

Table 8.3: Dataset for Pest Prediction, Detection & Identification (Part 2)

	Grass -hopper	Mites	Mosquito	Sawfly	Stem borer
Training (80%)	256	256	256	256	256
Testing (20%)	64	64	64	64	64

8.1.3 Weed Detection Identification :

The Cotton and Weed Dataset on Kaggle was created by Yuzhen Lu and contains approx. 400 images per class and are in .jpg format

Table 8.4: Dataset for Weed detection and Identification

	Carpet -weeds	Morning glory	Nutsedge	Palmer -Amaranth	Purslane	Waterhemp
Training (80%)	320	320	320	320	320	320
Testing (20%)	80	80	80	80	80	80

8.1.4 Species Recognition :

The PlantVillage dataset is a collection of images of various plant species, focusing on crop plants. The dataset contains approximately 1000 images per class, where each class corresponds to a particular plant disease or a healthy plant. The dataset is available in JPEG format and is commonly used for tasks such as plant disease detection and classification using machine learning algorithms. In this module, the focus is on the healthy images of plants from the PlantVillage dataset, which can be used for developing models to classify plants as healthy or diseased.

Table 8.5: Dataset for Species Recognition (Part 1)

	Apple	Blueberry	Cherry	Corn	Grape	Peach
Training (80%)	800	800	800	800	800	800
Testing (20%)	200	200	200	200	200	200

Table 8.6: Dataset for Species Recognition (Part 2)

	Pepper Bell	Potato	Raspberry	Soybean	Strawberry	Tomato
Training (80%)	800	800	800	800	800	800
Testing (20%)	200	200	200	200	200	200

8.1.5 Crop Recommendation :

The Crop Recommendation dataset created by Atharva Ingle on Kaggle contains data on various crop types. It consists of 8 columns and 2,700 rows. The dataset includes information such as temperature, humidity, rainfall, soil pH, soil nutrients, and crop type. This dataset can be used for crop recommendation systems that provide farmers with suggestions on what crops to plant based on environmental and soil conditions.

8.1.6 Crop Yield Prediction :

The Agriculture Crops Production in India dataset on Kaggle contains data on the production of 19 major crops in India from the year 1997-2015. The dataset has over 24,000 records, with each record representing a unique combination of crop, year, and season. The dataset provides information on the area harvested, production quantity, and yield of each crop. The data can be used to analyze the trends and patterns of crop production in India over the years and to identify the factors affecting crop production

8.2 Results Discussion and Performance Evaluation

8.2.1 Disease Detection and classification (For 2 classes):

Table 8.7: Disease detection and classification(for 2 classes)

Evaluation Parameter	CNN ResNet50	CNN DenseNet	Bins Approach (SVM)	Bins Approach (Decision Tree)	Bins Approach (Logistic regression)
Accuracy	96%	95%	90%	92%	95%
Precision	96%	96%	90%	92%	95%
Recall	96%	94%	90%	92%	95%
F1-Score	96%	95%	90%	92%	95%

Observation and Description: From Table 8.7, it can be observed that CNN with ResNet50 gives the best accuracy of 96% which is better than DenseNet and Bins Approach with Logistic Regression which both give accuracy of 95%. Bins with Decision tree gives an accuracy of 92% and the least accuracy is given by Bins with SVM which is 90%.

8.2.2 Disease Detection and classification (For 6 classes):

Table 8.8: Disease detection and classification(6 classes)

Evaluation Parameter	CNN ResNet50	CNN DenseNet	Bins Approach (SVM)	Bins Approach (Decision Tree)	Bins Approach (Logistic regression)
Accuracy	98%	95%	39%	54%	44%
Precision	98%	96%	35%	54%	46%
Recall	98%	94%	38%	54%	46%
F1-Score	98%	95%	32%	54%	43%

Observation and Description: Based on Table 8.8, it can be concluded that for six classes, the CNN model with ResNet50 architecture has the highest accuracy of 98%, which is superior to the DenseNet model with an accuracy of 96%. In contrast, the accuracy obtained by the Bins model with Decision tree and Logistic Regression are 54% and 46%, respectively, indicating a lower accuracy compared to the CNN models. The least accurate model among all the evaluated models is the Bins model with SVM, with an accuracy of only 39%.

8.2.3 Pest Detection and Classification:

Table 8.9: Pest detection and classification(9 classes)

Evaluation Parameter	CNN ResNet50	CNN DenseNet	Bins Approach (SVM)	Bins Approach (Decision Tree)	Bins Approach (Logistic regression)
Accuracy	99%	98%	14%	99%	21%
Precision	99%	98%	32%	99%	23%
Recall	99%	98%	13%	99%	21%
F1-Score	99%	98%	6%	99%	19%

Observation and Description: From Table 8.9, the accuracy of 99% was achieved by CNN with ResNet50 and Bins with Decision Tree, for the Pest Prediction module. The second best accuracy was given by CNN with DenseNet which is 98%. The performance of Bins with Logistic Regression and SVM is 21% and 14% respectively.

8.2.4 Weed Detection and Classification:

Table 8.10: Weed detection and classification(6 classes)

Evaluation Parameter	CNN ResNet50	CNN DenseNet
Accuracy	87%	95%
Precision	87%	95%
Recall	87.1%	95%
F1-Score	87%	95%

Observation and Description: From Table 8.10, it was observed that an accuracy of 87% was achieved by CNN with ResNet50 for the Weed Detection and Classification module. The best accuracy was given by CNN with DenseNet which is 95%. The performance of CNN with DenseNet was much better than CNN with ResNet50.

8.2.5 Species Recognition:

Table 8.11: Species Recognition(12 classes)

Evaluation Parameter	CNN ResNet50	CNN DenseNet	Bins Approach (SVM)	Bins Approach (Decision Tree)	Bins Approach (Logistic regression)
Accuracy	95%	96%	66%	79%	79%
Precision	95.2%	95%	67%	79%	78%
Recall	94.6%	96.1%	57%	81%	79%
F1-Score	95%	96%	58%	80%	78%

Observation and Description: From Table 8.11, it was observed that An accuracy of 96% was achieved by CNN with DenseNet for the Species Recognition module. The second best accuracy was given by CNN with ResNet50 which is 95%. Bins Approach gave relatively better accuracy than the other modules which was 66% for SVM, 79% for Decision Tree and 79% for Logistic Regression The performance of CNN with DenseNet was much better than CNN with ResNet50.

8.2.6 Crop Recommendation:

Table 8.12: Crop Recommendation

Evaluation Parameter	Naive Bayes	Decision Tree	Random Forest	SVM
Accuracy	97.1%	96.7%	98%	98%
Precision	97.7%	98.1%	98.6%	98%
Recall	97%	96.7%	98%	98%
F1-Score	96.9%	97.1%	98.1%	98%

Observation and Description: From Table 8.12, it can be observed from the evaluation parameters that Random Forest gives the best overall performance where the precision is 98.6% which is slightly better than that SVM which has precision of 98%. The performance of Naive Bayes and Decision tree is almost identical where the difference between the performances is negligible.

8.2.7 Crop Yield Prediction:

Table 8.13: Crop Yield Prediction

Evaluation Parameter	Random Forest Regressor	XGB Regressor	SVM
MAE	0.89	0.74	2.096
MSE	7.67	7.91	69.25
R2 Score	0.95	0.95	0.63

Observation and Description: Based on Table 8.13, it can be observed that the Random Forest Regressor and XGBRegressor models have a relatively similar performance in terms of MAE and MSE, with the XGBRegressor model performing slightly better in terms of MAE (0.74) and Random Forest Regressor performing slightly better in terms of MSE (7.67). However, the SVM model has a significantly higher MAE and MSE compared to both Random Forest Regressor and XGBRegressor models.

Chapter 9

Conclusion

The proposed "Precision Agriculture" was implemented successfully with the intention of supporting the farmers in various agricultural activities. It covered 6 modules: Disease Detection, Pest Detection, Weed Detection, Species Recognition, Crop Yield Prediction, and Crop Recommendation. For image detection and classification modules, image processing methods were focused more for image processing for extraction of features and machine learning techniques for final detection and classification. To achieve this, Bins approach as spatial domain image processing technique and CNN with ResNet50 and DenseNet as deep learning techniques have been used. Before applying these methods, preprocessing of the image dataset was done. Comparing the performance of all these approaches with respect to the performance evaluation parameters, we have found that CNN certainly plays an important role in feature engineering and is better than that of the Bins Approach. The overall comparison shows that CNN with ResNet50 is the best for all the image based modules except Species Recognition. Secondly, Bins Approach with Logistic Regression performs equally better than CNN with DenseNet that is 95% in Disease Detection and Classification. Observing the classifiers, Logistic Regression is found to be the best classifier for Bins Approach. The performance of Decision Tree Classifier and SVM classifier is decent compared to others. Based on our overall experiments and results, we can delineate the following:

1. Quicker and User-friendly: Classification systems are quicker than most, if not the best of all, traditional techniques. There is no need to go through lengthy procedures to get the results you want from the experts. The proposed system is straightforward, easy to implement and efficient.

2. Moreover, Bins approach with enhancement in feature extraction techniques can be explored as future work as it gives nearly equal results along with reduction to a great extent in the computational complexity. Furthermore, Bins approach with enhancement can be applied on multiple classes. Such an approach can be

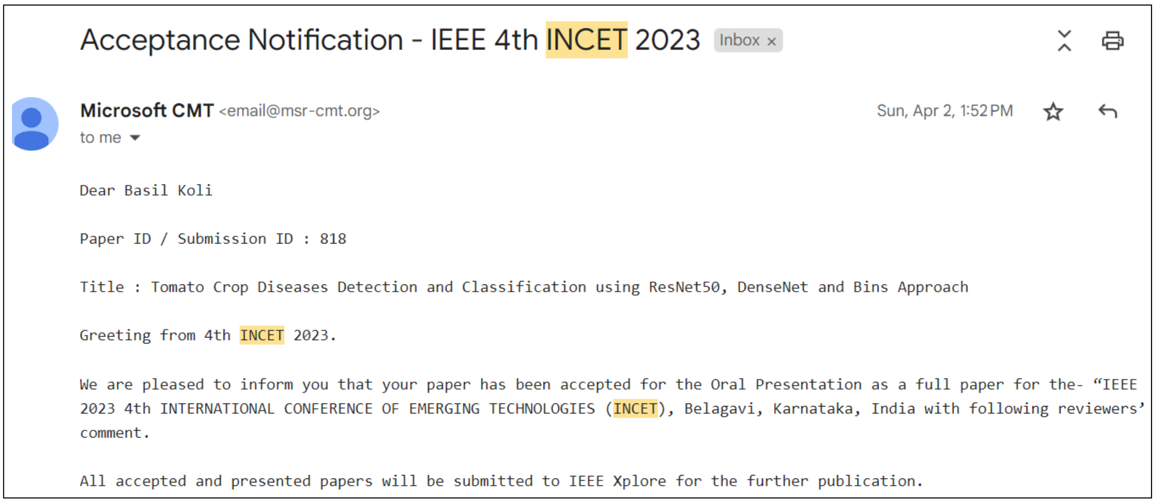
deployed in the agricultural domain, where immediate and proper diagnosis and treatment are of equal importance to have good crop health.

Based on the evaluation of different models for the Crop recommendation module, it can be concluded that Random Forest and SVM algorithms provide the best overall performance. The precision obtained by the Random Forest algorithm is 98.6%, which is slightly better than the SVM algorithm, which has a precision of 98%. However, the performance of Naive Bayes and Decision Tree algorithms is almost identical, with a negligible difference between their performances. Therefore, for the Crop recommendation module, both Random Forest and SVM algorithms can be considered as suitable choices. The selection of the final algorithm can be based on other factors such as computational resources, ease of implementation, and interpretability, among others.

For the Crop Yield Prediction module, it can be observed that the Random Forest Regressor and XGBRegressor models have a relatively similar performance in terms of MAE and MSE, with the XGBRegressor model performing slightly better in terms of MAE (0.74) and Random Forest Regressor performing slightly better in terms of MSE (7.67). However, the SVM model has a significantly higher MAE and MSE compared to both Random Forest Regressor and XGBRegressor models. In terms of R2 Score, both Random Forest Regressor and XGBRegressor models have a similar performance with a score of 0.95, while the SVM model has a significantly lower score of 0.63. Both Random Forest Regressor and XGBRegressor models have comparable performance in terms of MAE and MSE, with XGBRegressor performing slightly better in terms of MAE and Random Forest Regressor performing slightly better in terms of MSE. However, both models have a similar R2 Score of 0.95, indicating good fit to the data. The SVM model, on the other hand, has a significantly higher MAE and MSE and a lower R2 Score compared to the other models, suggesting a poorer fit to the data. Therefore, if the objective is to minimize MAE and MSE and maximize R2 Score, either Random Forest Regressor or XGBRegressor models can be considered as suitable choices.

Publications

Our paper titled **"Tomato Crop Diseases Detection and Classification using ResNet50, DenseNet and Bins Approach"** has been accepted in the **4th International Conference of Emerging Technologies 2023 (INCET 2023)** which is technically co-sponsored by **IEEE** Bangalore section. **INCET 2023** is indeed the premier conference of Asia Pacific for exchanging technical research in emerging technology. The conference aims to bring together researchers, engineers, and practitioners from academia and industry to share their experiences and insights on the latest developments and innovations in engineering and technology.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Dear Basil Koli

Paper ID / Submission ID : 818

Title : Tomato Crop Diseases Detection and Classification using ResNet50, DenseNet and Bins Approach

Greeting from 4th INCET 2023.

We are pleased to inform you that your paper has been accepted for the Oral Presentation as a full paper for the- "IEEE 2023 4th INTERNATIONAL CONFERENCE OF EMERGING TECHNOLOGIES (INCET), Belagavi, Karnataka, India with following reviewers' comment.

All accepted and presented papers will be submitted to IEEE Xplore for the further publication.

Figure 9.1: IEEE Paper Acceptance Email

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